Detecting AI-Generated Video: A Vision-Language Dual-View Survey

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The evolving realism of AI-generated Videos (AIGC-V) is rapidly rendering traditional artifact-centric detection insufficient, necessitating a paradigm shift from low-level inspection to high-level semantic verification. This paper presents a comprehensive survey of AIGC-V detection, reframing the task as Factual Fidelity Verification, which asks whether the events, entities, and physical processes depicted in a video are consistent with real-world facts. To systematize this rapidly evolving field, we propose a ***Vision-Language Dual-View*** taxonomy that organizes existing methods into a hierarchical, four-layer landscape, spanning intrinsic cue analysis, spatiotemporal consistency modeling, cross-modal consistency reasoning, and language-guided world-level reasoning. This dual-view framing highlights a fundamental transition from artifact matching in traditional deepfake detection to evidence-based semantic verification enabled by vision-language models and agentic reasoning pipelines. Based on a systematic review of 298 video-centered works, updated through July 2026, we synthesize AIGC-V generation paradigms, survey the landscape of detection methods, and review evaluation metrics and benchmarks in line with the proposed views. Finally, we discuss current challenges and identify promising directions toward robust, explainable, and trustworthy detection.

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Github: https://github.com/dxhou/AI-Generated-Video-Detection

Homepage: https://aigcvdetection.github.io/

1 Introduction

The rapid evolution of video generation models, exemplified by Sora 2 (OpenAI, 2024b), Veo 3 (Google DeepMind, 2025), and Seedance 2.0 (ByteDance Seed, 2026), is fundamentally reshaping the information landscape. Unlike early deepfakes dominated by localized face swapping (Wang et al., 2024b), modern **AI-Generated Content-Videos (AIGC-V)** have achieved cinematic fidelity with coherent narratives, increasingly blurring the

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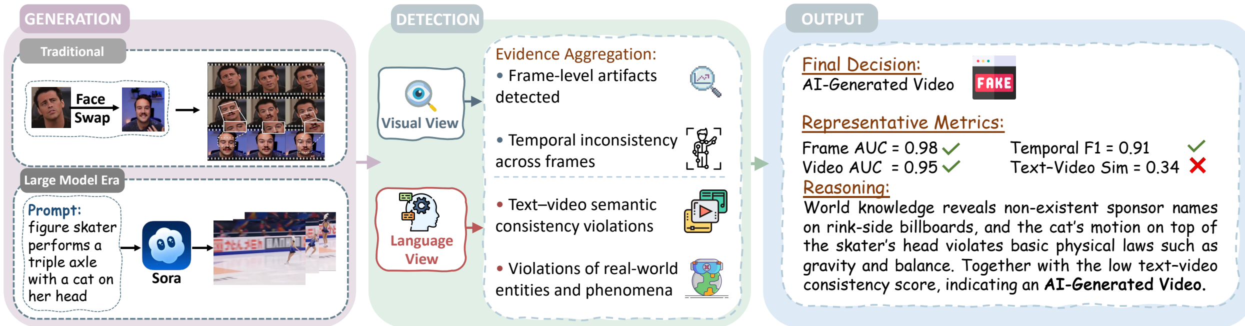

Figure 1 An example of the AIGC-V detection pipeline under our approach, illustrating AI-generated videos from traditional methods or text-to-video prompts, detection from the visual and language views, and outputs at different levels.

boundary between synthesized fiction and captured reality. This technological leap destabilizes the foundational trust in video evidence, traditionally relied upon to verify "who, when, where, and what" (Ho et al., 2022a).

As generation paradigms shift from local manipulation to end-to-end synthesis, such as text-to-video, traditional detection methods (Wang et al., 2025c; Ma et al., 2025) face a critical bottleneck. Early detection systems primarily relied on low-level visual artifacts, such as blending boundaries. However, advanced diffusion models and transformers (Ho et al., 2022b; OpenAI, 2024a) can now produce visually high-fidelity videos. This paradigm shift necessitates a new detection landscape, shifting from perceptual inspection (checking for visual artifacts) to cognitive reasoning (checking for semantic and factual violations). The surge of Vision-Language Models (VLMs)(Zhang et al., 2025d,c; Wang et al., 2026h) and agentic frameworks(Liu et al., 2024b, 2025g,f; Hou et al., 2024) offers a promising pathway, enabling detectors to perform world-level reasoning and verify cross-modal consistency. Although Fu et al. (2025) and Park et al. (2025) have begun to explore these directions, the existing literature remains fragmented. Meanwhile, surveys like Pei et al. (2024) and Liu et al. (2025c) are constrained to the early era of deepfakes or treat video merely as a sequence of images, failing to systematize the emerging class of methods that leverage language for semantic verification.

To address this gap, we propose a ***Vision-Language Dual-View*** taxonomy to organize the AIGC-V detection landscape, as illustrated in Figure 1. We reframe the problem as *Factual Fidelity Verification*, determining whether the video content aligns with real-world facts and physical laws, rather than simple binary classification. Our framework categorizes detection methods into a four-layer hierarchy, progressing from low-level perception to high-level cognition: (1) Intrinsic Cue Analysis, (2) Spatiotemporal Consistency, (3) Cross-Modal Consistency, and (4) Language-Guided World-Level Reasoning.

In this survey, we comprehensively review 298 video-centered works through July 2026. We begin with AIGC-V paradigms (§2), then formulate AIGC-V detection from the perspective of *factual fidelity* and introduce our Vision-Language dual-view framing (§3). We systematically organize methods under this landscape (§4), revisit evaluation metrics (§5.1) from a dual-view, and review benchmarks in line with AIGC-V paradigms (§5.2).

Finally, we identify the critical challenges (§6) in detecting increasingly sophisticated AIGC-V. By bridging the gap between traditional visual forensics and emerging multimodal reasoning, this paper aims to provide a structured roadmap for the next generation of explainable and trustworthy AIGC-V detection.

2 Paradigms of AI-Generated Video

Currently, diffusion-based and Transformer-based architectures are advancing rapidly in video generation, driving major progress in text-to-video and other AIGC-V pipelines (Ho et al., 2022b,a; Singer et al., 2022; OpenAI, 2024b; Google DeepMind, 2025; ByteDance Seed, 2026). In this survey, we group mainstream methodologies into three categories according to the underlying generation paradigm: local manipulation, audio-visual editing, and generative video synthesis, following representative systems in each regime (Brison et al., 2025; Cheng et al., 2022b; Ho et al., 2022b). Figure 2 provides an overview.

2.1 Local Manipulation

Methodologies categorized as *Local Manipulation* typically operate on authentic video sequences by modifying specific spatial regions or distinct semantic attributes (Brison et al., 2025; Heo and Woo, 2025). The resulting generation maintains a high degree of structural fidelity to the original footage. Another technical paradigm centers on video face swapping and facial element manipulation. Representative approaches typically utilize diffusion models (Wang et al., 2024b), 3D facial priors (Wang et al., 2025c), and controllable conditional encoding, integrating identity feature transfer with expression, pose, and illumination reconstruction from source videos into a unified generative framework. Given their ease of implementation and covert nature, these techniques represent a critical threat in real-world adversarial scenarios.

2.2 Audio-Visual Editing

Audio-Visual Editing uses speech as an explicit control signal to drive talking-head generation or to re-dub existing videos, requiring tight cross-modal alignment between audio and face dynamics while preserving identity and background (Cheng et al., 2022b). And recent diffusion-based or 3D-aware approaches (Ma et al., 2025; Hong et al., 2025) move toward unified conditional generation, where audio guidance and identity constraints are jointly modeled to produce lip-synced (Li et al., 2024a), visually consistent outputs. Due to their low operational barrier and strong perceptual plausibility, audio-visual edits can convincingly "bind" forged visual performances to plausible voice tracks, amplifying risks in impersonation and misinformation.

2.3 Generative Video Synthesis

Generative Video Synthesis targets end-to-end creation of complete video sequences from text or noise, fabricating both appearance and dynamics without authentic carriers.

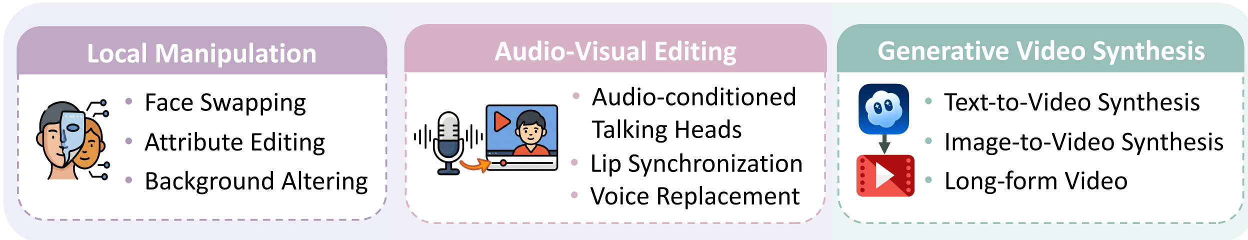

Figure 2 Overview of three AIGC-V paradigms defined in this survey, highlighting typical forms.

Diffusion-based generators extend image diffusion with temporal modeling for coherence, exemplified by Video Diffusion Models (Ho et al., 2022b) and cascaded high-fidelity pipelines such as Imagen Video (Ho et al., 2022a); Make-A-Video (Singer et al., 2022) further reduces reliance on paired text-video data by leveraging text-image priors and unpaired videos. Recent architectures further optimize representations: Show-1 (Zhang et al., 2025b) hybridizes pixel-based generation with latent refinement for efficiency, while Grid Diffusion (Lee et al., 2024) unifies spatiotemporal dimensions into a single 2D grid to simplify modeling. Transformer-style synthesis tokenizes videos and learns long-horizon structure via autoregressive or masked modeling, while DiT further highlights the scalability of Transformer backbones for diffusion-based generation (Peebles and Xie, 2023). At the industrial frontier, large-scale text-to-video systems such as Sora 2 (OpenAI, 2024b), Seedance 2.0 (ByteDance Seed, 2026), and Veo 3 (Google DeepMind, 2025) demonstrate strong open-domain capability and raise urgent needs for provenance and verification.

Where the Evidence Lives

- **LMV keeps a real carrier,** so the strongest clues are usually residual and spatially localized rather than scene-wide.
- **AVE is constrained by coupling,** making synchrony, speaker consistency, and speech-conditioned facial motion more decisive than generic image artifacts.
- **GVS removes the carrier altogether,** shifting detection away from local edit traces toward long-range coherence, factual verification, and provenance.

3 AI-Generated Video Detection

3.1 Task Scope

Visually high fidelity AIGC-V with richer narratives make the reliability of methods that output only a "real/fake" probability increasingly low in security and privacy scenarios (Wang et al., 2024a; Kaur et al., 2024), and the detection task should shift from merely answering the traditional question of "whether the video is AI generated" to **fact-level** judgment of whether the video content "objectively happened in the real world."

From a fact-level perspective, video content can be abstracted as a series of propositions of the form "at a certain time and place, which entities exist, and what happens to these entities." We define **Factual Fidelity** as "the fact-level propositions implied by the video and its metadata are consistent with the real world," and treat *Factual Fidelity Verification*

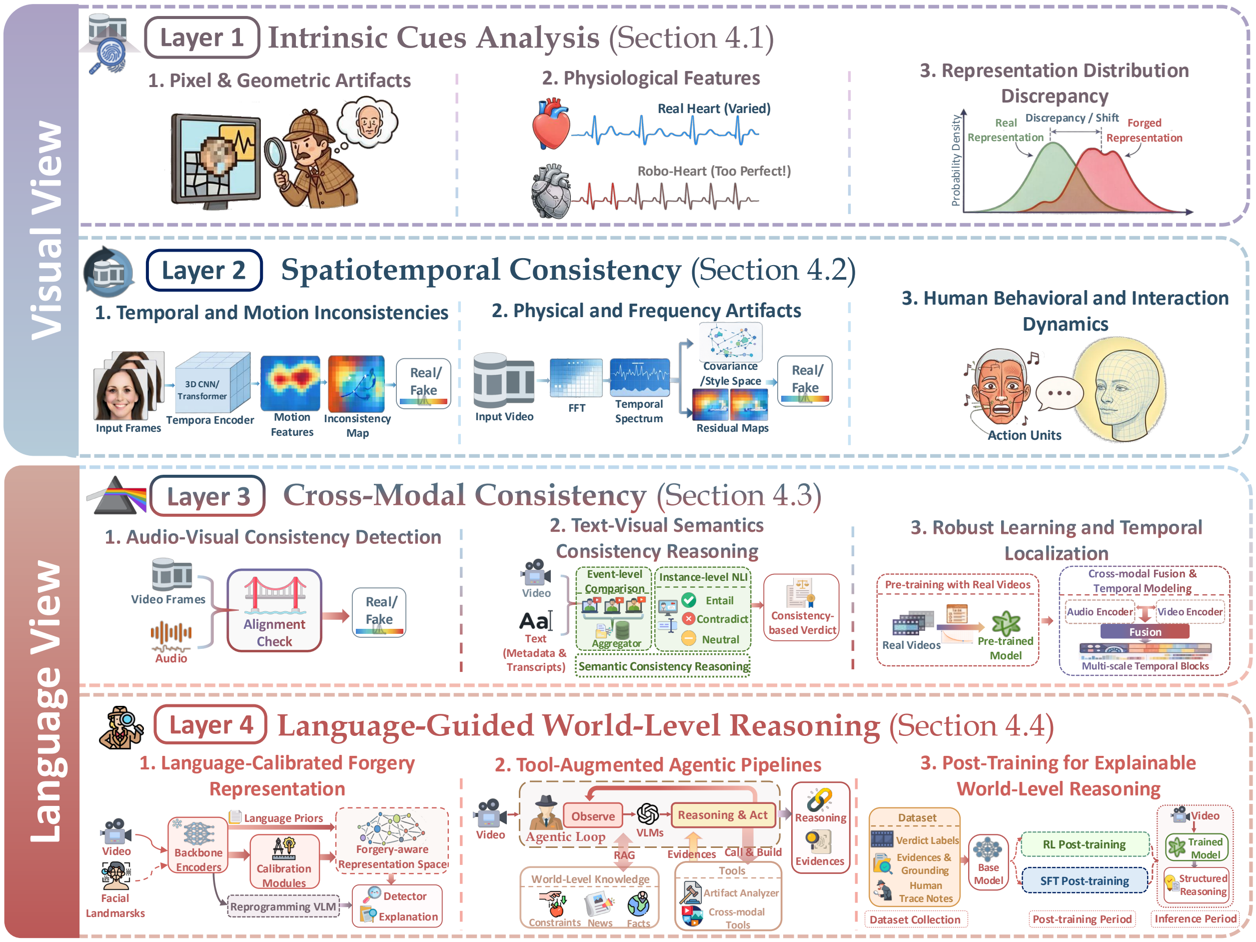

Figure 3 Dual-view, four-layer framework for AIGC-V detection. The visual view, spanning Layers 1-2, models intrinsic cues and spatiotemporal consistency, whereas the language view, spanning Layers 3-4, addresses cross-modal consistency and world-level reasoning, with each layer further decomposed into fine-grained subcategories. Boundary note: Layer 1 uses frame-level distributional evidence; Layer 2 models frame-sequence relations; Layer 3 performs within-video multimodal verification; Layer 4 requires external knowledge for verification.

as the **fundamental objective of AIGC-V detection**.

Corpus boundary. We include a work only when detecting, localizing, explaining, or benchmarking AI-generated or AI-manipulated *video* is a central objective. Audio-visual work qualifies when the analyzed object is a video clip, and a unified image–video method qualifies only when it gives video substantial method and evaluation treatment. We exclude image-first detectors and broad synthetic-media studies in which video is merely one minor modality. Layer assignment then follows the decisive evidence used by the detector, rather than title keywords or backbone names.

Furthermore, different evidence pathways provided by detection methods are integrated to indicate at which levels the video violates factual fidelity (Zhang et al., 2024b; Yu et al., 2025; Shen et al., 2025) and form an interpretable and trustworthy evidence space.

Definition 3.1 (Factual fidelity verification)**.** *Let $x \in \mathcal{X}$ be a video sample together with its metadata, and let $\Pi(x) \subseteq \mathcal{P}$ denote the set of fact-level propositions implied by x. Define the ideal factual-violation label as $y(x) = \mathbf{1}[\Pi(x) \nsubseteq \mathcal{P}_{\text{world}}]$, i.e., $y(x) = 1$ if and only if x violates factual fidelity.*

Definition 3.2 (AIGC-V detection)**.** *An AIGC-V detector is a mapping $D : \mathcal{X} \to [0,1] \times \mathcal{E}$. For a video sample $x \in \mathcal{X}$, $D(x)$ contains a* suspicion score $s(x) \in [0,1]$ *and* evidence $e(x) \in \mathcal{E}$ *supporting its judgment.*

We require $e(x)$ to be *trustworthy* and *grounded*. Concretely, we recommend representing $e(x)$ as a finite set of evidence items $e(x) = \{\epsilon_k\}_{k=1}^{K}$, where each $\epsilon_k = (g_k, c_k, \pi_k, r_k)$ contains: (a) a grounding g_k, such as temporal segment and optional spatial track, (b) a confidence $\pi_k \in [0,1]$, (c) a possible human-readable claim-check statement c_k describing what sub-claim is being verified and what inconsistency is found , and (d) a provenance record r_k such as tool outputs or retrieved facts enabling traceability.

3.2 Dual-view Four-Layer Taxonomy

The methodological perspectives of AIGC-V detection can be summarized as two complementary scientific views: visual and language. Out of this observation, as illustrated in Figure 3, we propose a **Vision-Language Dual-View** approach and organize existing methods into a four-layer methodological landscape from low-level perception to high-level cognition in §4.

Definition 3.3 (Multimodal video sample)**.** *Let $\mathcal{I} \subseteq \mathbb{R}^{H \times W \times C}$ denote the space of frames, and let $\mathcal{A}$, $\mathcal{S}$, $\mathcal{M}$ denote the spaces of audio, text, and metadata. A multimodal video sample is*

$$x = (V, A, S, M) \in \mathcal{X} := \mathcal{I}^T \times \mathcal{A} \times \mathcal{S} \times \mathcal{M},$$

where $V = (I_t)_{t=1}^{T}$ is the visual stream, and A, S, M may be empty when the corresponding modality is unavailable.

Visual View. The visual view focuses on the visual modality and emphasizes statistical differences between AIGC-V and real videos (Zhou et al., 2024; Gu et al., 2021). The detection pathway extends from low-level intrinsic cue analysis at the frame level in §4.1 to spatiotemporal consistency across frames in §4.2, forming the perceptual perspective for factual-fidelity verification.

Language View. In this survey, the language view denotes a *grounded verification pathway*, not an assumption that all audio cues are linguistic by default. Concretely, audio evidence enters this pathway in two forms. The first form is non-linguistic cross-modal perceptual evidence, which mainly checks within-video alignment without semantic parsing, such as lip–speech synchrony and onset and offset consistency (Chugh et al., 2020; Yang et al., 2021). The second form is language-grounded speech evidence: speech can be mapped via ASR or speech representations to phoneme, word, or semantic units, enabling consistency checks against identity and text content, such as voiceprint–identity coherence and speech–subtitle agreement (Shahzad et al., 2022; Bohacek and Farid, 2024; Cheng et al., 2022a; Zhang et al., 2025f). This design choice clarifies why audio can legitimately participate in the language view: it provides an interface from acoustic signals to language-level evidence and textual reasoning when needed. Building on these two cue types, methods progress from cross-modal consistency analysis (§4.3) to world-level reasoning at factual and knowledge levels when external evidence is required (§4.4) (Zhang et al., 2024b; Yu et al., 2025; Shen et al., 2025), thereby forming the cognitive perspective of factual-fidelity verification.

What AIGC-V Detection Must Establish

- **The target is factual fidelity, not only a real/fake score:** detection should judge whether the fact-level claims implied by a video and its metadata remain consistent with the real world.
- **A detector should return evidence, not only suspicion:** the output should couple a score with grounded, traceable evidence showing which claim is being checked, where the inconsistency lies, and why the decision is reliable.
- **Evidence can arrive through two complementary pathways:** the visual view tests intrinsic and temporal plausibility, while the language view moves from within-video cross-modal verification to explicit claim checking and world-level reasoning.

4 Landscape of Detection Methods

As illustrated in Figure 4, we organize AIGC-V detection methods into four layers, and provide a structured overview in this section. Our taxonomy is **evidence-oriented rather than task-exclusive**: layer assignment follows the dominant evidence pathway used for decision-making, not a hard partition of tasks. To make layer assignment operational, we use boundary criteria tied to evidence dependency: Layer 1 captures frame-level distributional cues that can be scored per frame and pooled, whereas Layer 2 requires explicit frame-sequence modeling of spatiotemporal relations. Layer 3 then moves from visual-only coherence to within-video multimodal verification, whereas Layer 4 further verifies claims against external facts, commonsense, or physics, even when within-video modalities appear mutually consistent. When a method combines cues from multiple layers, we map each module to its primary evidence type rather than forcing a single-task label. For each layer below, we further summarize the assumption-signal-failure-mode linkage to clarify why each pathway works, when it breaks, and how recent methods improve robustness under the factual-fidelity framing in Section 3. To make this trend concrete, Table 7 summarizes the video-first method corpus assigned in this section, using complete publication years through 2025 and a clearly marked partial-year snapshot through July 20, 2026; the four layer tables retain a compact representative subset for readability.

4.1 Layer 1: Intrinsic Cues Analysis

Layer 1 asks whether low-level visual signals obey the statistical regularities of real videos, whether they match acquisition and post-processing effects (Le and Woo, 2023; Wang et al., 2022b), and whether AI generation or editing leaves intrinsic cues such as style patterns, model artifacts (Chen et al., 2021), or deviant physiological signals (Ciftci et al., 2020b; Qi et al., 2020; Ciftci et al., 2020a). Methods operate from the visual view by modeling, extracting, and amplifying these low-level signals (Vahdati et al., 2024). The dominant

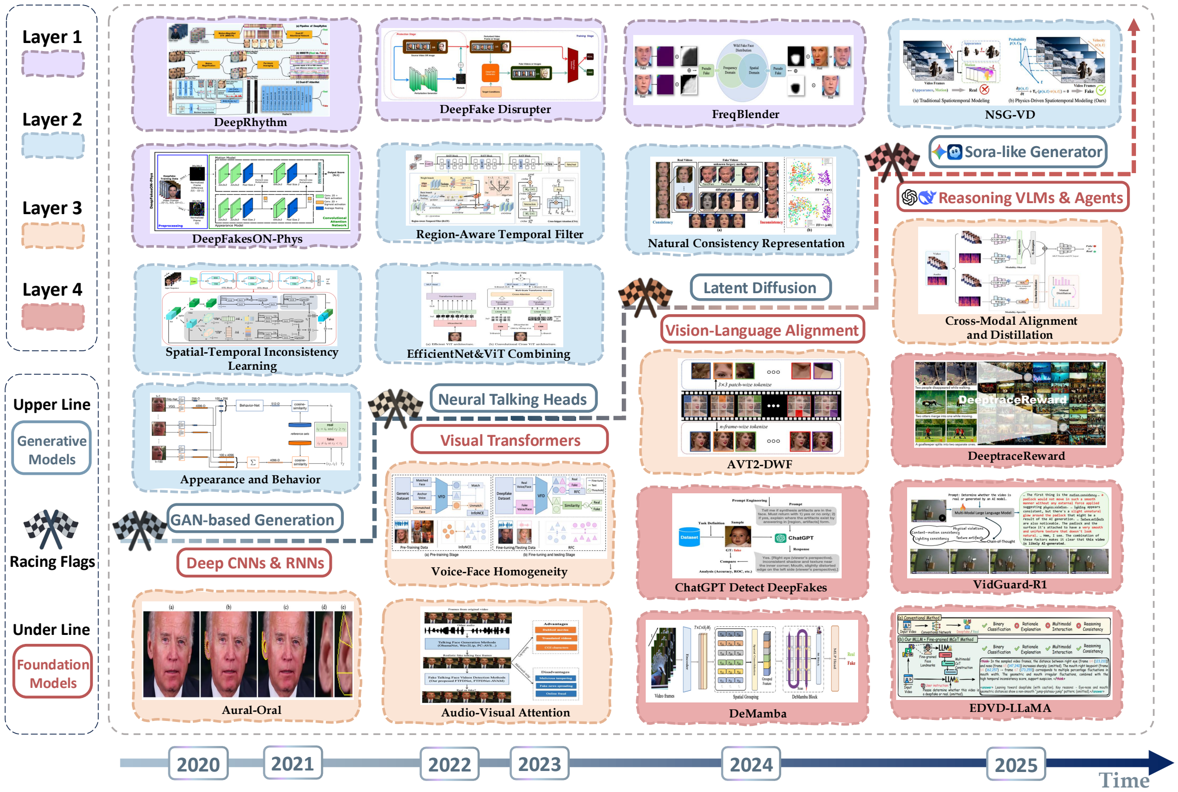

Figure 4 Landscape of representative AIGC-V detection methods aligned with the four-layer taxonomy in §4. Methods are grouped by the evolving capabilities of generative models that pose emerging threats and the corresponding advances in foundation models that enable detection. The year-wise shift toward language-view methods across the surveyed detection papers is quantified in Table 7.

evidence unit remains frame-level and distributional: a single frame can already yield an independent signal, while multiple frames mainly aggregate per-frame evidence. Once the decision depends on explicit inter-frame relations, the evidence moves to Layer 2. Table 1 summarizes representative methods in this layer.

Pixel and Geometric Artifacts. This line of work targets intrinsic traces introduced by AIGC-V synthesis and post-processing, including frequency fingerprints, texture residues, and geometric inconsistencies. FreqBlender amplifies synthesis-related spectral cues via spectral blending and contrastive learning (Zhou et al., 2024). SpecSem-Net similarly extracts high-frequency evidence but gates it with semantic context to suppress spectral noise; because its decision does not require explicit inter-frame relations, its decisive evidence remains Layer 1 (Wei et al., 2026b). Lightweight Complementary-Cue Fusion reaches the same layer boundary by combining wavelet-denoised appearance with phase-spectrum or local-binary-pattern channels before video-level aggregation (Baek et al., 2026). Such frame-wise frequency fingerprints remain Layer 1 evidence, whereas descriptors that depend on temporal-frequency behavior belong to Layer 2. Localized designs reduce reliance on global spectra by focusing on region-level residues, such as mask-guided localization in MagDR and texture-consistency learning in HCIL (Chen et al., 2021; Gu et al., 2022b). Geometric cues complement spectral and texture cues by testing structural plausibility, as instantiated by LRNet, RAM, and head-pose inconsistency tests (Sun et al., 2021; Tian et al., 2024; Yang et al., 2019). Following quality-aware training results (Le and Woo, 2023), these cues still degrade under compression and adaptive filtering in low-quality pipelines.

Physiological Features. Physiological cues treat faces as carriers of biological rhythms and micro-dynamics that are difficult to synthesize consistently. Because many practical AIGC-V scenarios remain human-centric, especially talking heads and avatars, we keep physiological evidence as a dedicated subclass. Recent follow-up is still limited: beyond classic rPPG pipelines, only a few works revisit local-global physiological interaction patterns (Wu et al., 2024; D'Amelio et al., 2023). Heart-related signals are extracted either by mapping residual representations to heart-rate domains (Ciftci et al., 2020b) or by learning spatiotemporal attention over rPPG-related patterns (Qi et al., 2020). Higher-order heart-rate variability and cross-region coupling improve robustness under noise (Hernandez-Ortega et al., 2020; Fernandes et al., 2019; Lee et al., 2021; Wu et al., 2024; Stefanov et al., 2022; Mao and Yang, 2021; Ciftci et al., 2020a). Complementary evidence includes blink irregularities and identity-specific muscle dynamics (Li et al., 2018; Cozzolino et al., 2021). These cues require stable facial visibility and sufficient temporal support, and remain sensitive to occlusion and aggressive re-encoding.

Representation Distribution Discrepancy. This line of work targets cross-generator transfer by modeling distribution gaps between authentic and synthetic videos rather than committing to a fixed artifact family. Latent and style-space perturbations diversify synthetic styles and reduce forgery specificity (Yan et al., 2024), while latent-flow modeling captures abnormal style trajectories (Choi et al., 2024). Curriculum learning and synthetic Blend-Fake data expose models to controlled difficulty and reduce shortcut learning (Lin et al., 2024b; Cheng et al., 2024). Recent follow-up stays close to low-level evidence: DFA combines foundation-model global priors with explicit local anomaly analysis to improve generalization under evolving forgery patterns (Liao et al., 2026). VINA treats video frames as physically grounded augmentations for a shared image–video decision boundary, while Real Data Lies shows that mismatched real/fake quality distributions can create a false impression of cross-generator generalization (Li et al., 2026d; Fang et al., 2026). G2VD attacks the same shortcut problem through controlled pixel- and frequency-domain counterfactual interventions followed by causal disentanglement (Du et al., 2026). GenD reaches strong cross-benchmark transfer by tuning only normalization parameters and learning on an L2-normalized feature manifold; its broad evaluation across deepfake video benchmarks also isolates the importance of paired real–fake source videos (Yermakov et al., 2026). DFD-HR adaptively prunes encoder depth, selects forgery-relevant tokens, and routes them through a mixture of experts, while QTFP purifies local forgery tokens and TSRL learns a sample curriculum from video-level performance changes; all three report video-level generalization even though their decisive representation remains frame-distribution based (Sun et al., 2026b; Wang et al., 2026a; Lei et al., 2026). Forensic Rewiring instead identifies shortcut-prone attention heads and selectively redirects them (A et al., 2026). Seeing the Unseen similarly targets detector transfer: it learns synthetic-video forensic microstructures from virtual 2D autoregressive generators, so its decisive signal remains an intrinsic visual trace rather than an explicit motion relation (Vahdati et al., 2026). CUPID instead uses person-of-interest reference videos: it reconstructs UV texture maps, matches query embeddings against pristine references, and exposes residual maps while keeping the decisive evidence identity- and appearance-based rather than sequence-relational (Affatato et al., 2026). Discrepancy objectives separate overlapping real and fake manifolds with bounded contrastive learning and explicitly modeled appearance shifts (Larue et al., 2023; Liu et al., 2024a). At inference time, one-shot adaptation (Chen et al., 2022), quality-aware training (Le and Woo, 2023), and locality-aware reconstruction (Du et al., 2020) improve robustness under shift while also surfacing region-level evidence for review.

Takeaways for Layer 1

- **Why It Works:** Low-level statistics still preserve residual capture and synthesis traces, so frame-wise forensic evidence can separate authentic from generated content.
- **Paradigm Fit:** Strongest in LMV; in AVE it is usually auxiliary unless the visual stream is also manipulated; in GVS, generator fingerprints help but transfer weakens across models, pipelines, and quality conditions.
- **Failure Modes:** Compression, transcoding, and domain shift attenuate these traces, and physiology-based cues are even more brittle because they require clear faces, stable signal quality, and favorable capture conditions.

Table 1 Layer 1: Intrinsic cues analysis. Representative methods grouped by cue family. *Cue* = evidence source; *Mechanism* = modeling design; Input: F=frames, V=video.

Method	Cue	Input	Mechanism	Output	Date
A. Pixel and geometric artifacts					
Inconsistent Head Poses (Yang et al., 2019)	3D head-pose inconsistency	V	Head-pose estimation + geometry checks	Score	05/2019
MagDR (Chen et al., 2021)	Localized manipulation artifacts	F+V	Mask-guided localization + reconstruction	Score+loc.	06/2021
HCIL (Gu et al., 2022b)	Region-level inconsistency	V	Hierarchical contrastive learning	Score	10/2022
NoiseDF (Wang and Chow, 2023)	Forensic noise (face vs. bg)	F	Denoised-noise fusion	Score	06/2023
FreqBlender (Zhou et al., 2024)	Spectral fingerprints	F	Spectral-blending augmentation	Score	12/2024
SpecSem-Net (Wei et al., 2026b)	Frame-level spectral artifacts	F	Spectral denoising + semantic gating	Score	05/2026
Lightweight Cue Fusion (Baek et al., 2026)	Wavelet, phase, texture	F	Handcrafted-cue fusion	Score	05/2026
B. Physiological features					
In Ictu Oculi (Li et al., 2018)	Eye-blink irregularities	V	Blink detection + temporal pattern modeling	Score	12/2018
FakeCatcher (Ciftci et al., 2020a)	PPG-like biological signal maps	V	Biological-map detector	Score	07/2020
Hearts (Ciftci et al., 2020b)	Heart signals in residuals	V	Residual maps → rPPG features	Score	09/2020
DeepRhythm (Qi et al., 2020)	Visual heartbeat rhythms (rPPG)	V	ST attention over rPPG	Score	10/2020
DeepFakesON-Phys (Hernandez-Ortega et al., 2020)	Heart-rate inconsistency	V	rPPG estimation + anomaly cues	Score	10/2020
Local rPPG Interaction (Wu et al., 2024)	Cross-region rPPG coupling	V	Local attention + long-range rPPG interaction	Score	02/2024
C. Distribution discrepancy and robustness					
OST (Chen et al., 2022)	Domain/compression shift	F	One-shot test-time adaptation	Score	12/2022
SeeABLE (Larue et al., 2023)	Soft shift discrepancies	F	Real-only bounded contrastive learning	Score	10/2023
QAD (Le and Woo, 2023)	Compression robustness	F	Quality-aware regularization	Score	10/2023
LSDA (Yan et al., 2024)	Cross-generator transfer	V	Latent augmentation for transfer	Score	06/2024
Style Latent Flows (Choi et al., 2024)	Abnormal style trajectories	V	Style-flow modeling + contrastive loss	Score	06/2024
Fake It Till You Make It (Lin et al., 2024b)	Curriculum forgery augmentation	V	Curriculum forgery augmentation	Score	10/2024
GenD (Yermakov et al., 2026)	Cross-benchmark shift	F+V	LN tuning + hyperspherical metric	Score	08/2025
DFA (Liao et al., 2026)	Global/local anomalies	F+V	CLIP adapter + local anomaly stream	Score	03/2026
VINA (Li et al., 2026d)	Image–video distribution shift	F	Natural-video augmentation + alignment	Score	05/2026
Real Data Lies (Fang et al., 2026)	Quality shortcuts	V	Quality matching + real expansion	Score	05/2026
DFD-HR (Sun et al., 2026b)	Layer/token forgery cues	F+V	Adaptive depth/token routing	Score	06/2026
QTFP (Wang et al., 2026a)	Local forgery-token evidence	F+V	Query-driven token purification	Score	06/2026
TSRL (Lei et al., 2026)	Learnable samples under shift	F+V	PPO tutor + dynamic curriculum	Score	06/2026
Forensic Rewiring (A et al., 2026)	Dataset-specific attention shortcuts	F+V	Head attribution + gradient masking	Score	06/2026
Seeing the Unseen (Vahdati et al., 2026)	Transferable microstructures	F+V	Virtual generators + trace learning	Score	06/2026
CUPID (Affatato et al., 2026)	POI UV-texture discrepancy	F+V	Reference matching + residuals	Score+loc.	06/2026
G2VD (Du et al., 2026)	Generator-specific shortcut cues	F	Counterfactual intervention + causal disentanglement	Score	07/2026

4.2 Layer 2: Spatiotemporal Consistency

Layer 2 asks whether spatiotemporal image flow satisfies the motion constraints of real videos. Real capture is constrained by continuous camera trajectories and physical scenes (Zhang et al., 2025e; Internò et al., 2025), so adjacent frames exhibit continuous, predictable, and physically feasible changes, whereas AIGC-V can show long-range inconsistencies such as object or background distortion (Gu et al., 2022c; Zhang et al., 2024a) and sudden local blurring (Gu et al., 2022a). Spatiotemporal inconsistency learning (Gu et al., 2021) and motion-based detection (Ma et al., 2024b; Yin et al., 2023; Haliassos et al., 2021) therefore exploit frame differences and flow residuals (Chen et al., 2025b). This makes Layer 2 distinct from Layer 1 by requiring explicit sequence modeling of inter-frame relations rather than frame-wise cue pooling. At the same time, Layer 2 remains a visual-modality layer: once verification requires semantic checks across audio, text, and visual events, the evidence pathway transitions to Layer 3. Table 2 summarizes representative methods in this layer.

Temporal and Motion Inconsistencies. Layer 2 makes explicit how temporal coherence is operationalized as a modeling prior. Instead of classifying frames independently, spatiotemporal methods model how facial appearance and geometry evolve across time and treat incoherence as evidence for forgery (Gu et al., 2021, 2022c). Clip encoders treat short sequences as space-time volumes and learn joint patterns with 3D CNNs or attention (Gu et al., 2021, 2022c; Hu et al., 2021; Zhang et al., 2024a; Chen et al., 2020; Zhang et al., 2021a). By jointly observing multiple frames, they reveal abnormal changes that are weak or ambiguous at the single-frame level. Region-aware variants highlight regions such as the eyes and mouth where features change abnormally, and auxiliary objectives can predict inconsistency maps to force localized temporal evidence (Gu et al., 2022c; Hu et al., 2021; Zhang et al., 2024a). Temporal Dropout and rhythm perturbation further reduce overfitting to specific frame rates or speaking rhythms (Chen et al., 2020; Zhang et al., 2021a). To reuse 2D backbones, some methods reshape frame sequences into grid-like layouts to enable efficient 2D processing and leverage image pre-training (Xu et al., 2024, 2023). Hybrid pipelines aggregate per-frame features with recurrent branches to fuse static appearance and dynamic motion cues at the decision stage (Masi et al., 2020). Application-oriented follow-ups continue to revisit CNN or LSTM aggregation as a lightweight temporal baseline for video deepfake detection (Bahadure and Mane, 2026). Motion-centric designs construct intermediate signals such as frame differences or optical flow to emphasize temporal transitions (Ma et al., 2024b; Yin et al., 2023). Flow-residual cues test whether local facial motion is compatible with global head motion, remaining informative under blur and compression (Chen et al., 2025b; Wang et al., 2023). Recent systems preserve and expose these relations in complementary ways. Native Scale avoids resizing away high-frequency and temporal evidence; ReConFuse aligns reconstruction errors with multi-frame semantics; and V-PVP replaces global readout with patch-velocity profiling (Li et al., 2026e; Chen et al., 2026b; Cui et al., 2026). CAST lets temporal tokens attend directly to spatial CNN evidence, while DFSL combines state-space and tri-frame temporal modules with Fourier–wavelet prompts; both explicitly bind fine spatial traces to their evolution over video (Thakre et al., 2026; Duan et al., 2026). Beyond Frames inflates CoAtNet's convolution and self-attention blocks into 3D operators over RGB or optical-flow clips, while 3D Differential Decomposition suppresses identity shortcuts by factorizing spatial and temporal derivatives into lightweight directional representations (Alattas et al., 2026; Gao et al., 2026). Color-Agnostic Detection couples temporal aggregation with a curriculum of color edits and color-histogram modulation to remain robust after common post-processing, whereas Spatiotemporal Defect Integration combines local frequency-amplified CNN evidence with long-range Transformer dynamics for joint video detection and generator attribution. SAGA extends this direction to multi-granular authenticity and generator attribution with interpretable temporal signatures (Zhu et al., 2026d; Liu et al., 2026a; Kundu et al., 2026). Training-free Over-coherence scores unusually redundant transitions of text-to-video generations in embedding space, while STALL jointly models spatial and temporal likelihood under real-video statistics and SVD-Det encodes raw RGB sequences, compression defects, and frame semantics with a lightweight 3D backbone (Brokman et al., 2026; Ben Hayun et al., 2026; Yang et al., 2026). SemaDet combines global CLIP semantics with a BiLSTM consistency branch, while FBAC-Net couples landmark-anchored features with bidirectional inference and Mamba–CNN temporal fusion to remain stable across compression levels (Xiang et al., 2026; Jiang et al., 2026). SPLIT measures patch-trajectory roughness and local motion incoherence without training, while LoCC detects lip-sync edits through counterfactual consistency between neighboring frames (Hyun and Kim, 2026; Datta et al., 2026). Despite its name, CAM-VFD fuses appearance, motion, and depth streams that all arise from the visual sequence; its decisive evidence is therefore Layer 2 rather than cross-modal Layer 3 (Elkhodary et al., 2026). DYMAPIA jointly models spatial, spectral, and optical-flow anomalies, whereas MAST uses spiking dynamics over temporal residuals together with semantic trajectories (Rana and Sung, 2026; Jang et al., 2026). A complementary systems direction spatially aggregates multiple frames into phase patterns and performs massively parallel video-level decoding with an optical-neural architecture (Kashani et al., 2026). FLARE analyzes short consecutive windows for flicker and unstable facial texture, then converts temporal evidence maps into spotlighted regions for a VLM explanation; because the VLM explains rather than changes the authenticity score, its decisive evidence remains Layer 2 (Panahi and de Sa, 2026). ASTAR Loss regularizes attention over frame sequences so that a video detector concentrates on temporally informative facial evidence rather than isolated frames (Al-rawahneh et al., 2026). Overall, spatiotemporal inconsistency methods frame deepfake detection as coherence analysis rather than static appearance classification.

Human Behavioral and Interaction Dynamics. Behavioral cues move from artifact matching to higher-level human realism, asking whether expressions, gaze, and identity evolve plausibly over time. Action-unit guided and appearance-behavior joint representations model muscle activations and expression trajectories to expose abnormal combinations or unnatural dynamics (Anand et al., 2025; Agarwal et al., 2020). Gaze-based cues probe attention and interaction patterns that are difficult to reproduce consistently in face swapping or reenactment (Kohler et al., 2025). Identity-driven methods test whether temporal dynamics remain compatible with the claimed person under varying pose, illumination, and compression (Dong et al., 2020). Geometry-consistency checks enforce compatibility between 2D observations and 3D face and head motion, exposing pose- or shape-inconsistent sequences (Sun et al., 2021; Yang et al., 2019; Tursman et al., 2020). BioLip sharpens this direction for lip-sync forgeries by modeling violations of landmark kinematics and biomechanical constraints across time, without relying on speech content; Forgery-Aware Lip likewise uses authentic-only lip dynamics, but shapes their distribution to reject unseen talking-face attacks (Chen and Xu, 2026; Chen et al., 2026a). Interpretable Facial Dynamics instead uses low-dimensional facial-movement trajectories and shows that higher-order temporal irregularities become especially discriminative during emotive expressions (Murphy et al., 2026). Facial Biometric Anomalies aggregates pairwise identity-embedding similarities over thousands of frames and detects implausibly large or small within-video identity variation without requiring a person-specific reference (Norman and Farid, 2026). These semantics- and geometry-aware cues can be strong against visually polished fakes but are sensitive to natural behavioral diversity and recording conditions.

Physical and Frequency Artifacts. Transform-domain designs compute descriptors where non-physical dynamics become separable from content variation, yielding traces that can be more stable than raw RGB. Compared with purely spatial or spatiotemporal backbones trained on RGB, these descriptors can be more robust to appearance changes and sometimes more transferable across generators (Kim et al., 2025). Temporal frequency responses expose abnormal periodic components and energy distributions along time, helping separate authentic motion spectra from synthetic artifacts (Nie et al., 2024; Kim et al., 2025). Physics-grounded authentication now also includes the Moiré motion invariant, which verifies the correlation between fringe phase and grating displacement in camera-captured video (Qing et al., 2026). Early-2026 work pushes this line toward higher-fidelity synthetic videos: MPF-Net combines static off-manifold deviation scoring with micro-temporal fluctuation analysis, arguing that visually polished clips can still reveal structured instability around the real-video manifold (He et al., 2026). Second-order statistics capture covariance mismatches and subtle distribution shifts in spatiotemporal features that survive mild post-processing (Zheng et al., 2025; Padhi et al., 2025). Residual- and difference-based features amplify deviations from real-video manifolds by emphasizing temporal changes rather than absolute appearance (Xu et al., 2025d; Stamnas and Sanchez, 2025). Noise Amplification extends this idea to bit-plane signals with pixel-, region-, and frame-level aggregation, whereas STREAM operates directly on compressed-domain I-frames, motion vectors, and residuals for efficient detection (Cheng et al., 2026; Zhu et al., 2026a). Event-guided interpolation detection combines event-derived temporal change with high-frequency frame artifacts, while VIDS-Guard fuses SRM, YCbCr, and Fourier residual streams before temporal-transformer aggregation over face clips (Guo et al., 2026; Alanazi and Asif, 2026). Aletheia injects optical-flow curl, specular-reflectance statistics, and rPPG spectra into localized artifact attention, while SpInShield adversarially perturbs temporal spectra to suppress fragile frequency shortcuts (Ghori, 2026; Gu et al., 2026a). Physics-inspired priors provide complementary constraints on motion feasibility and perceptual plausibility, linking detection to camera trajectories and physical scenes (Zhang et al., 2025e; Internò et al., 2025). Two representative detectors in this subclass are NSG-VD (Zhang et al., 2025e) and ReStraV (Internò et al., 2025). NSG-VD injects explicit physics constraints into spatiotemporal modeling by matching normalized spatiotemporal gradients and distribution statistics between real and generated videos, while ReStraV scores how far latent motion trajectories deviate from straight, natural perceptual manifolds.

Generalization and Robustness. Beyond cue design, robustness strategies aim to maintain performance under unseen generators and post-processing by changing objectives, architectures, and training protocols (Nguyen et al., 2025; Yan et al., 2025; Trinh et al., 2021; Coccomini et al., 2022; Tong et al., 2025). Vulnerability-aware objectives and plug-and-play adaptation modules steer backbones toward transferable spatiotemporal evidence (Nguyen et al., 2025; Yan et al., 2025). Unified or cross-model detectors reuse pretrained backbones and expand the hypothesis space to handle diverse manipulations and media types (Trinh et al., 2021; Veeramachaneni et al., 2025; Coccomini et al., 2022; Liu et al., 2025a; Tariq et al., 2021). Forensic augmentation and quality-centric curricula expand coverage of perturbations and hard cases (Corvi et al., 2025; Song et al., 2024a). Compression-aware scalable detection jointly predicts authenticity and MPEG degradation with multi-head or multi-branch designs, reducing the need to train a separate detector for each compression level (Perelli et al., 2026). Self-supervised video forensics learns normal dynamics via synchronization or anomaly objectives and then treats deviations as manipulations (Yang et al., 2025; Feng et al., 2023). Multiple-instance formulations aggregate clip-level evidence to reduce reliance on exhaustive forged labels and handle mixed-quality videos (Li et al., 2020a).

Takeaways for Layer 2

- **Why It Works:** Real capture follows continuous trajectories and feasible motion, whereas generated video still accumulates subtle long-horizon temporal inconsistencies.
- **Paradigm Fit:** Especially useful in GVS and in LMV when edits disturb motion across frames; in AVE, it matters mainly when the visual stream itself is synthesized, such as lip reenactment.
- **Failure Modes:** Short clips, resampling, and compression dilute the signal, and detectors may still collapse to frame-level shortcuts instead of genuine temporal reasoning.

Table 2 Layer 2: Spatiotemporal consistency. Representative methods grouped by (A) temporal and motion inconsistencies, (B) physical and frequency artifacts, and (C) human behavioral and interaction dynamics in §4.2. *Cue* = evidence source; *Mechanism* = modeling design. All methods take video as input.

Method	Cue	Mechanism	Output	Date
A. Temporal and motion inconsistencies				
FS-Spotter (Chen et al., 2020)	Spatiotemporal swap artifacts	Fuse spatial+temporal cues	Score	07/2020
Dynamic Prototypes (Trinh et al., 2021)	Transferable spatiotemporal evidence	Dynamic prototypes + predictive learning	Score	01/2021
Temporal Dropout 3DCNN (Zhang et al., 2021a)	Frame-rate dependent temporal artifacts	3D CNN + temporal dropout	Score	08/2021
STIL (Gu et al., 2021)	Short-clip spatiotemporal inconsistency	Clip encoder for ST inconsistency	Score	10/2021
EfficientNet-ViT Ensemble (Coccomini et al., 2022)	Robust temporal features	CNN and ViT ensemble	Score	05/2022
Region-Aware Temporal Inconsistency (Gu et al., 2022c)	Region-wise dynamics anomalies	Region-aware learning + maps	Score+loc.	07/2022
TALL (Xu et al., 2023)	Cross-frame inconsistency	Thumbnail layout + 2D backbone	Score	10/2023
Natural Consistency Representation (Zhang et al., 2024a)	Temporal naturalness deviations	SSL consistency + fine-tune	Score	11/2024
Vulnerability-Aware Learning (Nguyen et al., 2025)	Cross-generator vulnerability patterns	Vulnerability-aware transfer	Score	01/2025
GC-ConsFlow (Chen et al., 2025b)	Optical-flow residual inconsistency	Flow residuals + global context	Score	06/2025
Plug-and-Play Adapters (Yan et al., 2025)	Generalization under generator shift	Video blending + ST adapters	Score	06/2025
CAST (Thakre et al., 2026)	Coupled spatial and temporal artifacts	CNN–Transformer cross-attention fusion	Score	06/2025
SemaDet (Xiang et al., 2026)	Global semantics + inter-frame consistency	CLIP branch + BiLSTM temporal branch	Score	11/2025
DFSL (Duan et al., 2026)	Dual-frequency and tri-frame dynamics	State-space modules + Fourier–wavelet prompts	Score	12/2025
Beyond Frames (3D-CoAtNet) (Alattas et al., 2026)	Multi-frame RGB and optical-flow dynamics	Inflated 3D CoAtNet clip encoder	Score	02/2026
3D Differential Decomposition (Gao et al., 2026)	Directional spatial–temporal derivatives	Identity-suppressed differential factorization	Score	02/2026
Compression-Aware Scalable Detection (Perelli et al., 2026)	Authenticity cues under MPEG degradation	Multi-head/branch compression supervision	Score	02/2026
Over-coherence (Brokman et al., 2026)	Unnatural temporal embedding coherence	Training-free transition criterion	Score	03/2026
STALL (Ben Hayun et al., 2026)	Spatial–temporal likelihood under real data	Training-free probabilistic scoring	Score	03/2026
SVD-Det (Yang et al., 2026)	RGB, compression, and semantic dynamics	3D-Swin + hierarchical DoQA fusion	Score	03/2026
Native Scale (Li et al., 2026e)	Preserved native-scale ST artifacts	Variable-resolution Qwen2.5-VL video encoder	Score	04/2026
DYMAPIA (Rana and Sung, 2026)	Spatial, spectral, and flow anomalies	Dynamic masks + lightweight classifier	Score+loc.	04/2026
FBAC-Net (Jiang et al., 2026)	Compression-robust facial dynamics	Landmark anchors + bidirectional Mamba–CNN	Score	04/2026
MAST (Jang et al., 2026)	Residual spikes + semantic trajectories	SNN temporal branch + frozen encoder	Score	05/2026
CAM-VFD (Elkhodary et al., 2026)	Appearance–motion–depth contradiction	Cross-attention over visual feature streams	Score	05/2026
Optical-Neural Detector (Kashani et al., 2026)	Multi-frame spatial/Fourier encoding	Digital encoder + optical decoder	Score	05/2026
ReConFuse (Chen et al., 2026b)	Temporally organized reconstruction errors	WF-VAE errors + semantics + Mamba	Score	06/2026
FLARE (Panahi and de Sa, 2026)	Short-window flicker and texture instability	Temporal detector + evidence-guided explanation	Score+exp.	06/2026
LoCC (Datta et al., 2026)	Neighbor-frame lip-motion consistency	Counterfactual frame comparison	Score+loc.	06/2026
Color-Agnostic Detection (Zhu et al., 2026d)	Color-edit robust temporal evidence	Contrastive curriculum + color-aware modulation	Score	06/2026
Spatiotemporal Defect Integration (Liu et al., 2026a)	Local frequency traces + long-range dynamics	CNN–Transformer fusion + joint attribution	Score+attr.	06/2026
SAGA (Kundu et al., 2026)	Multi-granular generator traces	Video transformer + pretrain-and-attribute	Score+attr.	06/2026
SPLIT (Hyun and Kim, 2026)	Patch roughness + motion incoherence	Training-free frozen-encoder scoring	Score+loc.	07/2026
V-PVP (Cui et al., 2026)	Local patch-velocity profiles	Velocity-gated dual-stream readout	Score	07/2026
ASTAR Loss (Alrawahneh et al., 2026)	Temporally informative facial regions	Feature-guided temporal attention regularization	Score	07/2026
B. Physical and frequency artifacts				
DIP (Nie et al., 2024)	Temporal frequency response anomalies	Transform-domain temporal spectra	Score	12/2024
ReStraV (Perceptual Straightening) (Internò et al., 2025)	Non-physical motion and trajectory irregularities	Perceptual straightening	Score	07/2025
Beyond RGB (Kim et al., 2025)	Stable descriptors beyond raw RGB	Transformed descriptors	Score	07/2025
Physics-Driven ST Modeling (Zhang et al., 2025e)	Physics constraints on spatiotemporal flow	Physics-driven ST modeling	Score	10/2025
MPF-Net (He et al., 2026)	Off-manifold residuals + micro-temporal fluctuations	Hierarchical manifold deviation + temporal filtering	Score	01/2026
Moiré Authentication (Qing et al., 2026)	Camera-induced optical motion invariant	Fringe-phase/displacement correlation	Score	04/2026
Aletheia (Ghori, 2026)	Flow, reflectance, and rPPG physics	Physics-conditioned artifact attention	Score+loc.	04/2026
Event-Guided DVFI Detection (Guo et al., 2026)	Interpolation events + high-frequency artifacts	Event-guided temporal and spectral fusion	Score	04/2026
SpInShield (Gu et al., 2026a)	Temporal-spectrum shortcut attacks	Learnable spectral adversary + suppression	Score	05/2026
Noise Amplification (Cheng et al., 2026)	Bit-plane detail and temporal noise	Pixel/region/frame amplification	Score	06/2026
STREAM (Zhu et al., 2026a)	Compressed-domain coding artifacts	I-frame + motion-vector + residual mining	Score	05/2026
VIDS-Guard (Alanazi and Asif, 2026)	Spatial, color, and spectral residuals	Three-stream encoder + temporal transformer	Score	05/2026
C. Human behavioral and interaction dynamics				
Emotions Don't Lie (Mittal et al., 2020)	Affective audio-visual cue coherence	Siamese audio-visual model + triplet loss	Score	03/2020
Appearance and Behavior (Agarwal et al., 2020)	Appearance and behavior realism over time	Behavior-aware temporal features	Score	12/2020
Identity-Driven Detection (Dong et al., 2020)	Identity dynamics consistency	Track identity dynamics	Score	12/2020
Gaze Tracking (Kohler et al., 2025)	Gaze dynamics	Gaze features for dyadic calls	Score	09/2025
Facial Biometric Anomalies (Norman and Farid, 2026)	Within-video identity-embedding variation	Pairwise biometric statistics + XGBoost	Score	07/2025
BioLip (Chen and Xu, 2026)	Lip-landmark biomechanical violations	Kinematic constraint modeling	Score	04/2026
Forgery-Aware Lip (Chen et al., 2026a)	Authentic lip dynamics under unseen attacks	Real-only contrastive distribution shaping	Score	06/2026
Interpretable Facial Dynamics (Murphy et al., 2026)	Higher-order facial-motion irregularities	Behavioral trajectories + interpretable ML	Score	04/2026

4.3 Layer 3: Cross-Modal Consistency

Layer 3 focuses on whether the modalities in a video are well aligned in describing the same thing. Real videos often contain audio, text, and visuals that are highly aligned across modalities, whereas AIGC-V may present mismatches between lip movements and speech or between identity and voiceprint (Chugh et al., 2020; Yang et al., 2021; Zhang et al., 2021b; Cheng et al., 2022a). Mismatches can also appear between image content and associated text, motivating text-video consistency reasoning (Zhang et al., 2025f; Liu et al., 2025e; Wang et al., 2025a). Symbolic and semantic lip-sync analyses further enlarge this consistency space (Shahzad et al., 2022; Bohacek and Farid, 2024; Li et al., 2024b; Koutlis and Papadopoulos, 2025). Compared with Layer 2, the key shift is from visual-only sequence coherence to within-video multimodal verification; compared with Layer 4, evidence in Layer 3 is still primarily within-video and does not require external world knowledge as a mandatory source.

To clarify the role of audio in this layer, our taxonomy does not treat all audio cues as linguistic by default. Instead, we distinguish two cue types that naturally coexist in Layer 3. The first type is non-linguistic cross-modal perceptual cues, which test alignment without semantic parsing, such as lip-speech synchrony and onset and offset consistency. The second type is language- grounded speech cues, where speech is mapped via ASR or speech representations into phoneme, word, or semantic units, enabling explicit verification against identity and text content, such as voiceprint-identity coherence and speech-subtitle agreement, and providing a language interface for further reasoning when external knowledge is needed. Table 3 summarizes representative methods in this layer.

Audio-Visual Consistency Detection. Audio-visual consistency methods decompose evidence into synchrony, symbolic alignment, and identity coherence. Synchrony cues compare speech dynamics with mouth motion and learn cross-modal embeddings that penalize misalignment (Chugh et al., 2020; Yang et al., 2021). Cues extend beyond the mouth to additional regions whose micro-motions correlate with speech production and head dynamics (Agarwal and Farid, 2021). Symbolic alignment improves interpretability by mapping audio and visual streams to token or phoneme sequences and comparing them at the sequence level (Shahzad et al., 2022; Bohacek and Farid, 2024; Li et al., 2024b; Koutlis and Papadopoulos, 2025). TISAN makes this hierarchy explicit by comparing lip-reading and ASR transcripts at the word level before applying real-only, cross-attentive phoneme alignment to unseen lip-sync attacks (He and Wang, 2026). Identity coherence shifts the question from “what is said” to “who is speaking” by testing whether voice and face embeddings belong to the same person, including person-of-interest settings trained with authentic talking-head data only (Zhang et al., 2021b; Cheng et al., 2022a; Cozzolino et al., 2023; Muppalla et al., 2023). ConLLM further couples contrastive multimodal alignment with LLM-based reasoning to reduce modality fragmentation and expose fine-grained inconsistencies that are difficult to separate with shallow fusion alone (Kashyap et al., 2026). Generator-side probes such as X-AVDT further exploit inversion-accessed cross-attention signals to expose hidden speech-motion correspondence errors, while AV-LMMDetect shows that large multimodal models can also be directly fine-tuned as within-video audio-visual verifiers without requiring external world knowledge (Kim et al., 2026;

Cao et al., 2026b). ICASSP 2026 systems make the fusion policy more explicit: HybridMask learns complementary coarse spatiotemporal and fine facial masks before cross-attention, whereas AVATAR uses reinforcement learning to assign per-clip audio and visual fusion weights (Cao et al., 2026a; Shabbir et al., 2026). Recent video-first work expands both the training signal and the interaction setting. MSCT uses differential cross-modal attention, and AVPF synthesizes pseudo-fakes from authentic clips to reduce dependence on forgery labels (Wei et al., 2026a; Wei and Li, 2026). Listening Deepfake Detection shifts the target from speaking to reactive listener motion, while From Talking to Singing exposes distinct rhythm and articulation failures in synthetic singing heads (Liu et al., 2026c,b). Attribution-Guided Multimodal Detection regularizes the shared embedding with generator attribution and aligns audio–visual forensic fingerprints, while EMO-BOOST models intra- and inter-modal emotion consistency as a complementary high-level cue (Ahmad et al., 2026; Marik et al., 2026). The Semantic Mismatch study further shows that authentic audio and authentic video can still form a forged claim when their meanings diverge, extending four-way modality-integrity tests with an explicit real-audio/real-video mismatch class (Deshmukh et al., 2026). The CVPR 2026 self-supervised representation study systematically probes audio, visual, and joint audio–visual encoders with linear and anomaly-detection heads, finding complementary forensic information but persistent cross-dataset failures (Boldisor et al., 2026). EAV-DFD addresses the same transfer gap with an audio, visual, and joint audio–visual ensemble whose teacher–student adaptation uses a small target-domain video subset while retaining modality-specific predictions (Abolhasani et al., 2026). Recent journal and preprint systems further separate distinct sources of cross-modal failure. Predictive Inter-Modal Alignment reconstructs audio–visual associations under asynchronous capture, FoVB factorizes modality-specific and shared forgery variables, and AVSCNet decouples synchronization from content consistency (Wang et al., 2026g; Nie et al., 2026; Zhu et al., 2026b). HAVIC learns structural and micro/macro coherence priors from authentic videos, AVIR removes redundant local and cross-modal information, and AMID disentangles temporal, spatial, and semantic interactions before adaptive routing (Peng et al., 2026; Liu et al., 2026d; Xue et al., 2026). VATS combines fast temporal alignment with interpretable manipulation localization, while Emotion-Aware Detection uses facial, vocal, and textual emotion agreement as a few-shot generalization signal (Zha et al., 2026; Zhang et al., 2026b).

Text-Video Semantic Consistency Reasoning. Text-video semantic consistency reasoning reframes detection as claim verification: given a video and its associated text, such as titles, captions, or OCR and ASR transcripts, the model tests whether the textual statements are supported by visual evidence, with semantic contradictions serving as a key signal. CA-FVD supervises cross-modal consistency by prompting a VLM to obtain pairwise modality-consistency pseudo labels across visual, text, and audio streams, then training with cosine-based consistency losses; co-attention fusion and collaborative diagnosis aggregate modality-level confidence into a final score (Zhang et al., 2025f). CSCL moves from global scoring to grounded evidence by building patch-token consistency matrices and using cascaded decoders to separately model within-modality context and cross-modality semantics; forgery-aware aggregation reduces confusing but locally consistent content and enables localization (Liu et al., 2025e). To handle event shifts, T^3SVFND introduces test-time training with a reconstruction objective conditioned on multimodal context, suggesting event-robust text-video reasoning for detection settings (Wang et al., 2025a). Multimodality Semantic Consistency Fusion samples a video sequence into time-indexed visual observations and associated text, then gates cross-attention by image–text semantic agreement while retaining event-level weak supervision for detection and localization (Sun et al., 2026a). For fully generated video, ATSS constructs visual, textual, and cross-modal temporal self-similarity matrices from frame descriptions, whereas CMTA measures the temporal stability of visual–textual alignment at coarse and fine granularities (Wang et al., 2026c,b).

Robust Learning and Temporal Localization. In realistic settings, cross-modal inconsistencies are often temporally sparse, so localization is needed not only for final scoring but also for review and diagnosis. Robust methods therefore need to remain stable under compression, language shift, and partial-stream corruption while still marking when and where the mismatch emerges. This couples robustness with interpretability: a useful detector should preserve discrimination under noise while exposing the specific temporal span that carries the cross-modal failure rather than diffusing evidence over the whole clip. Two-stream fusion and attention-based alignment provide a common backbone for joint modeling and interpretation (Zhou and Lim, 2021; Wang et al., 2022a). Transferable representations learned from large-scale authentic data improve stability under compression and language shifts (Liang et al., 2025; Feng et al., 2023). Weak supervision with video-level labels and bias-aware objectives enables fine-grained temporal localization without dense annotations (Xu et al., 2025c; Astrid et al., 2025). AuViRe reconstructs speech representations in each direction across audio and lip motion, using reconstruction discrepancies to localize manipulated intervals (Koutlis and Papadopoulos, 2026). Audio shift prediction provides an additional self-supervised proxy for sparse cross-modal disturbance and improves temporal localization in multimodal deepfake settings (Anshul et al., 2026). Challenge-driven systems such as Divide and Conquer (Li et al., 2026a) also show the value of decomposing detection into audio-side and visual-side localization before late fusion, which is particularly useful when manipulations are sparse or concentrated in only one stream. The latest localizers make this evidence more explicit: a joint LMM, spatiotemporal visual, and partial-spoof audio representation supports detection and boundary prediction; IaMSB suppresses noise transfer under single-sided or asynchronous forgeries; GEM-TFL converts clip labels into latent modality attributes before graph-based proposal refinement; MG-RWKV adapts temporal receptive fields with linear-complexity sequence modeling; and EVAS progressively couples audio–visual evidence before calibrating sparse forgery boundaries (Le et al., 2026; Xiong et al., 2026a; Zhu et al., 2026e; Ni et al., 2026; Shen et al., 2026). CCFormer instead cascades visual suspicious-segment screening with cross-modal frame-level refinement, turning binary filtering into precise audio–visual boundary localization (Li and Ren, 2026).

Takeaways for Layer 3

- **Why It Works:** Authentic videos obey strong within-video coupling across modalities, so mismatches in timing, identity, or semantics are hard to hide consistently.
- **Paradigm Fit:** Natural fit for AVE; useful in LMV when edits change speaker identity or event semantics and usable audio or text is present; in GVS, it is strongest for captioned or prompt-conditioned clips.
- **Failure Modes:** Its value drops when audio or text is missing, noisy, or weakly informative, and attackers can partially restore apparent alignment through redubbing, retiming, or other resynchronization.

Table 3 Layer 3A: Audio-visual consistency. Representative methods based on cross-modal alignment and coherence.

Method	Cue	Input	Mechanism	Output	Date
A. Audio-visual consistency detection					
Not Made for Each Other (Chugh et al., 2020)	Audio-visual dissonance	Speech+Face	Cross-modal mismatch + localization	Score+loc.	10/2020
Dynamic Lip Movement (Yang et al., 2021)	Lip motion vs. speech mismatch	Speech+Lip	Lip-motion dynamics analysis	Score	12/2020
Aural-Oral Dynamics (Agarwal and Farid, 2021)	Aural-oral dynamics coherence	Speech+Lip	Audio+mouth dynamics features	Score	06/2021
Joint Audio-Visual Detection (Zhou and Lim, 2021)	Audio-visual synchrony and semantics	Speech+Face	Joint synchrony+semantic modeling	Score	10/2021
Audio-Visual Attention (Wang et al., 2022a)	Cross-modal alignment	Speech+Face	Cross-attention fusion	Score	03/2022
Lip Sync Matters (Shahzad et al., 2022)	Symbolic lip-sync mismatch	Speech+Lip	Token and phoneme sequence comparison	Score	11/2022
Voice-Face Homogeneity (Cheng et al., 2022a)	Voice-face identity coherence	Voice+Face-ID	Compare voice and face-ID embeddings	Score	11/2023
Lost in Translation (Bohacek and Farid, 2024)	Language-aware audio-visual mismatch	Speech+Lip	Language-conditioned lip-sync	Score	06/2024
AVFF (Oorloff et al., 2024)	Intrinsic audio-visual correspondences	Speech+Face	Real-only SSL pretrain + classifier	Score	06/2024
Multi-task Audio-Visual Prompt Learning (Miao et al., 2025)	Fine-grained audio-visual consistency	Speech+Face	Prompting + matching loss + fusion	Score	04/2025
CAD (Du et al., 2025)	Semantic audio-visual misalignment + cues	Speech+Face	Alignment + distillation	Score	05/2025
FauForensics (Wang et al., 2026d)	Facial-action and audio–visual coherence	Speech+Face	Action-unit-guided multimodal fusion	Score	05/2025
Predictive Inter-Modal Alignment (Wang et al., 2026g)	Audio–visual asynchrony and missing links	Audio+Video	Subspace alignment + masked reconstruction	Score	09/2025
PIA (Datta et al., 2025)	Phoneme-timing mismatch + ID dynamics	Speech+Lip	Phoneme timing + ID dynamics	Score	10/2025
KLASSify to Verify (Kukanov and Ng, 2025)	Robust audio-visual cues (unseen attacks)	Speech+Face	Audio SSL + graph attn + handcrafted	Score+loc.	10/2025
Audio-Visual SSL Study (Boldisor et al., 2026)	Complementary pretrained AV representations	Audio+Video	Linear/anomaly probes + representation fusion	Score+loc.	11/2025
FoVB (Nie et al., 2026)	Modality-specific and shared forgery evidence	Audio+Video	Variational Bayes + orthogonal factorization	Score	11/2025
Emotion-Aware Detection (Zhang et al., 2026b)	Facial, vocal, and textual emotion mismatch	A/V/T	Contrastive learning + text-guided fusion	Score	01/2026
TISAN (He and Wang, 2026)	Word and phoneme audio–visual mismatch	Speech+Lip	Transcript check + real-only alignment	Score	01/2026
ConLLM (Kashyap et al., 2026)	Multimodal semantic inconsistency	Speech+Face	Contrastive alignment + LLM reasoning	Score	01/2026
AV-LMMDetect (Cao et al., 2026b)	Within-video audio–visual inconsistency	Audio+Video	Audio/video tokens + multimodal LLM tuning	Score	02/2026
AVSCNet (Zhu et al., 2026b)	Synchronization and content inconsistency	Audio+Video	Dual branches + hierarchical cross-attention	Score	02/2026
X-AVDT (Kim et al., 2026)	Generator-internal audio-visual correspondence	Speech+Face	DDIM inversion + cross-attention probing	Score	03/2026
VATS (Zha et al., 2026)	Fast alignment and modality-specific artifacts	Audio+Video	Multitask Transformer + localization head	Score+loc.	03/2026
HAVIC (Peng et al., 2026)	Intra- and inter-modal intrinsic coherence	Audio+Video	Real-only coherence priors + adaptive fusion	Score	03/2026
MSCT (Wei et al., 2026a)	Differential audio-visual discrepancy	Speech+Face	Differential cross-modal attention	Score	04/2026
HybridMask (Cao et al., 2026a)	Coarse ST + fine facial AV mismatch	Speech+Face	Complementary masks + cross-attention fusion	Score	04/2026
AVATAR (Shabbir et al., 2026)	Clip-adaptive audio–visual reliability	Audio+Video	Dual encoders + PPO-weighted fusion	Score	04/2026
AVPF (Wei and Li, 2026)	Self-generated audio-visual mismatch	Speech+Face	Authentic-only pseudo-fake learning	Score	04/2026
Listening Deepfake Detection (Liu et al., 2026c)	Listener motion vs. speaker audio	Audio+Listener	Motion-aware audio-guided fusion	Score	04/2026
Attribution-Guided Multimodal (Ahmad et al., 2026)	Cross-modal forensic fingerprints	Speech+Face	Detection–attribution regularization	Score+attr.	04/2026
Semantic Mismatch (Deshmukh et al., 2026)	Audio–video semantic divergence	Speech+Face	Five-class mismatch modeling + ImageBind	Score	04/2026
Talking-to-Singing (Liu et al., 2026b)	Singing articulation and rhythm mismatch	Singing+Face	Audio-visual temporal modeling	Score	05/2026
EMO-BOOST (Marik et al., 2026)	Cross-modal emotion consistency	Speech+Face	Emotion dynamics + detector fusion	Score	05/2026
EAV-DFD (Abolhasani et al., 2026)	Cross-domain audio-visual evidence	Audio+Video	Ensemble + teacher–student adaptation	Score+mod.	06/2026
AVIR (Liu et al., 2026d)	Redundant/noisy audio–visual evidence	Audio+Video	Superpixels + information bottleneck	Score	05/2026
AMID (Xue et al., 2026)	Temporal, spatial, and semantic AV interactions	Audio+Video	Cross-modal reconstruction + adaptive LoRA	Score	07/2026

Table 4 Layer 3B: Text–video semantic consistency. Representative methods for semantic agreement and temporal reasoning.

Method	Cue	Input	Mechanism	Output	Date
B. Text-video semantic consistency reasoning					
Semantic Consistency Fusion (Sun et al., 2026a)	Image–text semantic agreement over time	Video+Text	Consistency-gated cross-attention	Score+loc.	01/2026
CA-FVD (Zhang et al., 2025f)	Video-text semantic mismatch	Video+Text	VLM pseudo-labels + consistency loss	Score	04/2025
CSCL (Liu et al., 2025e)	Text-image inconsistency grounding	Frame+Text	Cascaded consistency decoders	Score+loc.	06/2025
T^{3}SVFND (Wang et al., 2025a)	Event-shifted semantics	Video+Text	Test-time MLM reconstruction	Score	07/2025
ATSS (Wang et al., 2026c)	Visual/textual temporal self-similarity	Video+Text	Triple-similarity matrices + fusion	Score	04/2026
CMTA (Wang et al., 2026b)	Visual–textual alignment stability	Video+Text	Captioning + multi-grained temporal modeling	Score	05/2026

Table 5 Layer 3C: Robust learning and temporal localization. Representative cross-modal methods emphasizing robustness or localized evidence.

Method	Cue	Input	Mechanism	Output	Date
C. Robust learning and temporal localization					
Cross- and Within-Modality Regularization (Zou et al., 2024)	Modality separation under perturb.	Speech+Face	Audio-visual Transformer + cross and within regs	Score	04/2024
Audio-Visual Local Inconsistencies (Astrid et al., 2025)	Local temporal audio-visual inconsistency	Speech+Face	Local inconsistency + localization	Score+loc.	04/2025
Circumventing Shortcuts (Smeu et al., 2025)	Shortcut-robust audio-visual reps (silence)	Speech+Face	Real-only SSL audio-visual alignment	Score	06/2025
SpeechForensics (Liang et al., 2025)	Real-only audio-visual repr. learning	Speech+Lip	SSL pretrain on real + fine-tune	Score	08/2025
WMMT (Xu et al., 2025b)	Weakly-sup. temporal loc.	Speech+Face	Multitask + MoE + deviation loss	Score+loc.	08/2025
HOLA (Wu et al., 2025)	Hierarchical audio-visual aggregation	Speech+Face	SSL pretrain + gated aggregation	Score	10/2025
Weakly-Supervised Temporal Localization (Xu et al., 2025c)	Sparse multimodal deviations	Speech+Face	Weak supervision + deviation modeling	Score+loc.	10/2025
AuViRe (Koutlis and Papadopoulos, 2026)	Cross-modal speech reconstruction error	Speech+Lip	Bidirectional AV reconstruction	Score+loc.	11/2025
CCFormer (Li and Ren, 2026)	Sparse audio–visual forgery boundaries	Audio+Video	Cascaded screening + cross-modal refinement	Score+loc.	12/2025
A-V Shift Prediction (Anshul et al., 2026)	Temporal audio-visual perturbation cues	Speech+Face	Audio shift prediction + localization	Score+loc.	01/2026
Divide and Conquer (Li et al., 2026a)	Cross-modal fusion + temporal localization	Speech+Face	Unimodal detectors + score fusion	Score+loc.	02/2026
Multi-Modal Forgery Representation (Le et al., 2026)	Semantic, visual, and audio forgery cues	Video+Audio	LMM + ST visual + partial-spoof audio	Score+loc.	05/2026
IaMSB (Xiong et al., 2026a)	Asynchronous and single-sided forgeries	Speech+Face	Inconsistency-aware Schrödinger bridge	Loc.	05/2026
GEM-TFL (Zhu et al., 2026e)	Weakly supervised forgery intervals	Audio+Video	EM latent attributes + graph refinement	Loc.	06/2026
MG-RWKV (Ni et al., 2026)	Multi-duration temporal boundaries	Speech+Face	Bidirectional RWKV + granularity MoE	Loc.	07/2026
EVAS (Shen et al., 2026)	Sparse audio–visual forgery boundaries	Audio+Video	Multi-stage AV synergy + boundary calibration	Loc.	07/2026

4.4 Layer 4: Language-Guided World-Level Reasoning

Layer 4 elevates detection from internal consistency within the video to consistency with world-level rules and knowledge. The research question shifts to whether the video content can truly exist in the real world and remain reasonable in semantic and factual dimensions. Real videos should be consistent with real-world facts, physical rules, domain knowledge, and basic common sense, whereas AIGC-V often fails to align fully with these constraints, creating the detection space exploited by this layer (Motamed et al., 2025b). Language serves as a natural interface for human-readable explanations, while baseline VLM prompting in zero-shot or few-shot settings provides a minimal route for detecting inconsistencies and producing textual rationales (Jia et al., 2024; Shahzad et al., 2025). Compared with Layer 3, the detector is no longer asked only whether modalities agree, but whether the implied claim remains tenable once external facts, commonsense, or physics are brought into the loop. Table 6 summarizes representative methods in this layer.

Language-Calibrated Forgery Representations. Language-calibrated designs inject prompts or textual priors into multimodal representations to reshape forgery-sensitive features without rebuilding the backbone. CPML aligns physiological pulse (rPPG) signals and facial landmark dynamics with prompts and enforces cross-quality and cross-modal consistency to stabilize physiological evidence under compression (Lai et al., 2024; D'Amelio et al., 2023). RepDFD reprograms pretrained vision-language models by freezing the backbone and learning input-side perturbations and adaptive prompts, while knowledge-guided variants build textual prototypes and uncertainty modeling for improved transfer (Lin et al., 2025a; Yu et al., 2025; Shen et al., 2025). VLAForge preserves pretrained vision–language alignment while coupling a forgery perceiver with an identity-aware, text-prompted alignment score, making language semantics part of the authenticity decision rather than an explanation-only add-on (Zhu et al., 2026c). VFT-VLM similarly injects an explicit texture, edge, pixel, and frequency trace encoder into InternVL2, then aligns this forensic evidence with textual explanations through lightweight adaptation (Wang et al., 2026f). Adapter-style reprogramming provides a lightweight alternative when full fine-tuning is infeasible (Shao et al., 2025). These methods are useful when external retrieval is unavailable but language priors still need to bind visual cues to explicit semantic hypotheses and human-readable defect categories.

Evidence-Guided and Agentic Pipelines. Evidence-guided pipelines cast AIGC-V detection as explicit evidence gathering, where language guides what to inspect and, in tool-augmented settings, which analyzer to call next. LAVID uses an observe-tool-integrate loop: it forms a coarse hypothesis, calls external tools, updates prompts based on intermediate outputs, and fuses multi-step evidence (Liu et al., 2025d). FakeHunter adds memory-anchored observe-think-act routines, including retrieval of semantic anchors and reuse of intermediate results across steps (Chen et al., 2025a). DAVID-XR1 and related pipelines emphasize semantic anchors and modular analyzers to support interpretable evidence composition (Gao et al., 2025). Multi-agent coordination further decomposes modality-specific evidence before fusion (Zaman et al., 2025). The attraction of this design is that the final verdict can be traced to retrieved facts, tool outputs, and intermediate judgments rather than a single opaque forward pass. The 2026 wave makes this design more selective and auditable. DVAR uses adversarial multi-agent debate; CoCoDetect invokes MLLM reasoning only for uncertain samples; MSLoc pairs segment localization with natural-language explanation; and VMD-FACT represents multimodal evidence and fact-checking dependencies as an interpretable graph (Qi et al., 2026; Feng et al., 2026a,b; Zhang et al., 2026c). SafeGuard separates perception from social-risk reasoning, while Hermes combines instance-conditioned retrieval, an evidence reasoning graph, forensic tools, and multi-agent deliberation (Wu et al., 2026; Li et al., 2026b). Artifact-Bench complements these systems with explicit evaluation of both authenticity decisions and artifact diagnosis (Tang et al., 2026).

Post-Training for Explainable World-Level Reasoning. Post-training internalizes evidence selection and reasoning style so that inference depends less on external controllers. Within Layer 4, reinforcement learning and preference alignment encode how evidence should be gathered and how explanations should be structured, as explored by VidGuard-R1, Veritas, and BusterX++ (Park et al., 2025; Tan et al., 2025; Wen et al., 2025b). DeeptraceReward learns complementary reward supervision aligned to human-perceived fakeness in AI-generated videos (Fu et al., 2025), and VideoVeritas further couples preference alignment with perception pretext reinforcement learning and fact-based reasoning for more balanced AIGC-V verification (Tan et al., 2026). Related efforts extend post-training to multi-stage reasoning and modular explanation under complex AIGC-V settings (Wen et al., 2025a; Sun et al., 2025; Li et al., 2026c; Chen et al., 2024b). The emphasis shifts from generic instruction-following to disciplined justification, including when to retrieve, when to localize uncertainty, and how to present compact but auditable evidence.

Method	Base Model and System	Training	Output	Date
A. Prompts and adapters for representation calibration				
ChatGPT Detect (Jia et al., 2024; Shahzad et al., 2025)	ChatGPT	Prompt	Verdict+exp.	06/2024
CPML (Lai et al., 2024)	rPPG+landmark encoder	Prompt-guided	Score	11/2024
DeepFake-Adapter (Shao et al., 2025)	ViT (frozen)	Adapter tuning	Frame label (agg.)	01/2025
RepDFD (Lin et al., 2025a)	Frozen VLM	Prompt tuning	Label	04/2025
AuthGuard (Shen et al., 2025)	Vision encoder + LLM	Cls+ITC; uncertainty	Verdict+exp.	06/2025
LVLMDFD (Yu et al., 2025)	LVLM + LLM	Detector+prompt learner	Verdict+loc.+exp.	07/2025
VLAForge (Zhu et al., 2026c)	VLM + ForgePerceiver	Identity-aware text prompting + VLA score	Score	03/2026
VFT-VLM (Wang et al., 2026f)	InternVL2 + forgery-trace encoder	LoRA + trace–language alignment	Verdict+exp.	03/2026
B. Evidence-guided and agentic systems				
LAVID (Liu et al., 2025d)	LVLM agent + tools	Tool loop	Verdict+evidence	02/2025
DAVID-XR1 (Gao et al., 2025)	Video VLM	SFT	Loc.+defect reasoning	06/2025
FakeHunter (Chen et al., 2025a)	Vision+audio encoders + VLM agent	Agent+retrieval	Verdict+evidence	08/2025
DeepAgent (Zaman et al., 2025)	Multi-agent pipeline	Not stated	MM verdict	12/2025
DVAR (Qi et al., 2026)	Video MLLM agents	Multi-agent debate	Verdict+debate	04/2026
CoCoDetect (Feng et al., 2026a)	Detector + MLLM gate	Uncertainty-guided selective reasoning	Verdict+evidence	05/2026
MSLoc / TASLE (Feng et al., 2026b)	Long-video forensic model	Segment localization + explanation	Loc.+exp.	06/2026
VMD-FACT / IEEG (Zhang et al., 2026c)	MLLM + evidence graph	Fact checks + dependency reasoning	Verdict+graph	06/2026
SafeGuard (Wu et al., 2026)	Perception and reasoning agents	Multi-agent social-risk verification	Verdict+exp.	07/2026
Hermes (Li et al., 2026b)	RAG + forensic tools + agents	Evidence graph + deliberation	Verdict+evidence	05/2026
C. Post-training, preferences and rewards				
X2-DFD (Chen et al., 2024b)	LLaVA + aux detectors	SFT (explainable data)	Verdict+exp.	10/2024
BusterX (Wen et al., 2025a)	Qwen2.5-VL	SFT → RL	Verdict+exp.	05/2025
BusterX++ (Wen et al., 2025b)	Qwen2.5-VL	RL → SFT → RL	Verdict+struct. exp.	07/2025
VERITAS (Tan et al., 2025)	InternVL3 and Qwen2.5-VL	SFT → MiPO → P-GRPO	Verdict+exp.	08/2025
DeeptraceReward (Fu et al., 2025)	VideoLLaMA3 and Qwen2.5-VL with reward modeling	Reward-model train	Reward	09/2025
VidGuard-R1 (Park et al., 2025)	Qwen2.5-VL	SFT → DPO → GRPO	Verdict+exp.	10/2025
EDVD-LLaMA (Sun et al., 2025)	Qwen2.5-7B + video encoder	SFT	Verdict+exp.	10/2025
Skyra and Skyra-RL (Li et al., 2026c)	Qwen2.5-VL	SFT → RL	Verdict+grounded reasoning	12/2025
VideoVeritas (Tan et al., 2026)	Video MLLM	Pref. align + PPRL	Verdict+reasoning	02/2026

Table 6 Layer 4: Language-guided world-level reasoning. Representative methods grouped by (A) prompts and adapters, (B) evidence-guided and agentic systems, and (C) post-training with preferences and rewards. *Base Model and System* reports backbone names when available.

Takeaways for Layer 4

- **Why It Works:** It targets a harder bottleneck than within-video agreement: a clip may look self-consistent yet still fail once its implied claims are checked against facts, commonsense, causality, or physical rules.
- **Paradigm Fit:** Best for GVS and claim-centric LMV or AVE cases where the decision turns on external verification of actors, events, relations, or physical plausibility, especially when within-video modalities appear mutually consistent.
- **Failure Modes:** Verification depends on knowledge coverage, retrieval freshness, and correct claim decomposition, and VLMs can hallucinate, over-abstract, or miss subtle visual evidence while sounding confident.

4.5 Quantitative Summary of the Four-Layer Landscape

4.5.1 Year-wise Layer Distribution

Table 7 summarizes the video-first method corpus assigned to the four layers in Section 4 and tracks how the taxonomy evolves over time. The layer tables display a compact representative subset of this corpus, while the counts also cover the additional methods discussed in the surrounding text. We report both yearly share and cumulative share. The yearly ratio reflects release-time dynamics within each publication year, whereas the cumulative ratio shows how the full surveyed landscape shifts as methods accumulate. Because 2026 is still in progress, the table marks its July 20 snapshot with an asterisk; the cumulative denominator additionally includes the three surveyed papers before 2020.

Year	Layer Counts					View Counts		Language-View Share	
	L1	L2	L3	L4	Total	Visual	Language	Yearly	Cumulative
2020	5	7	1	0	13	**12**	**1**	**7.7%** 1 of 13	**6.3%** 1 of 16
2021	5	6	4	0	15	**11**	**4**	**26.7%** 4 of 15	**16.1%** 5 of 31
2022	4	3	2	0	9	**7**	**2**	**22.2%** 2 of 9	**17.5%** 7 of 40
2023	3	3	4	0	10	**6**	**4**	**40.0%** 4 of 10	**22.0%** 11 of 50
2024	9	5	4	3	21	**14**	**7**	**33.3%** 7 of 21	**25.4%** 18 of 71
2025	1	22	20	15	58	**23**	**35**	**60.3%** 35 of 58	**41.1%** 53 of 129
2026*	12	32	30	9	83	**44**	**39**	**47.0%** 39 of 83	**43.4%** 92 of 212
Δ	*+7*	*+25*	*+29*	*+9*	*+70*	*+32*	*+38*	*+39.3*	*+37.1*

Table 7 Year-wise quantitative summary for the video-first method corpus surveyed in Section 4; the layer tables show its compact representative subset. The visual view combines Layers 1 and 2, whereas the language view combines Layers 3 and 4. Share values are reported as a percentage with the corresponding count ratio. 2026* is a partial-year snapshot through July 20; the cumulative denominator additionally includes the three surveyed papers before 2020. In the Δ row, differences are computed between 2020 and the partial 2026 snapshot, and the two share columns report percentage-point changes.

Two patterns stand out. First, the early landscape is still overwhelmingly visual-view: from 2020 to 2022, Layers 1 and 2 account for 30 of 37 surveyed methods, showing that the field was primarily organized around forensic traces and temporal coherence. Second, the transition toward the language view is gradual rather than monotonic. The yearly language-view share rises from 7.7% in 2020 to 26.7% and 22.2% in 2021–2022, reaches 40.0% in 2023, and then expands further in 2024 and 2025. The decisive break arrives in 2025, when 35 of 58 surveyed methods fall into the language view. The cumulative language-view share correspondingly grows from 6.3% to 41.1%, indicating that multimodal verification and world-level reasoning have moved from a marginal line of work to a major axis of recent AIGC-V detection. At the same time, the visual view remains substantial, with 23 methods still belonging to Layers 1 and 2 in 2025; low-level and temporal evidence are therefore not replaced, but increasingly complemented by higher-level verification. The partial 2026 snapshot remains closely balanced: 39 of 83 surveyed methods belong to the language view, while 44 rely primarily on intrinsic or spatiotemporal visual evidence. The cumulative language-view share reaches 43.4% (92 of 212) by July 20, 2026.

4.5.2 Layer-wise Performance Snapshot

To complement the taxonomy with a compact quantitative anchor, Table 8 collects reported AUC (%) for representative methods across the four layers. Because reporting protocols remain heterogeneous, each number is paired with its protocol label, either cross-dataset (CD) or in-domain (ID). The table should therefore be read as a protocol-aware snapshot rather than as a single cross-layer leaderboard.

Method	Layer	Protocol	CDFv2	DFDCP	DFDC
Layer 1					
FreqBlender (Zhou et al., 2024)	L1	CD	94.6	87.6	74.6
SeeABLE (Larue et al., 2023)	L1	CD	87.3	86.3	75.9
LSDA (Yan et al., 2024)	L1	CD	83.0	81.5	73.6
Style Latent Flows (Choi et al., 2024)	L1	CD	89.0	–	–
Layer 2					
TALL-Swin (Xu et al., 2023)	L2	CD	90.8	–	76.8
LipForensics (Haliassos et al., 2021)	L2	CD	82.4	–	73.5
Two-branch (Masi et al., 2020)	L2	CD	76.6	–	–
Layer 3					
MDS (Chugh et al., 2020)	L3	ID	–	–	90.6
Layer 4					
RepDFD (Lin et al., 2025a)	L4	CD	89.9	95.0	81.0
LVLMDFD (Yu et al., 2025)	L4	CD	94.3	92.4	77.0

Table 8 AUC (%) snapshot across the four-layer taxonomy with explicit protocol labels. CD = cross-dataset with training on FF++ and testing on CDFv2, DFDCP, and DFDC; ID = in-domain. CD and ID values are **not directly comparable**, so the table should be read as a protocol-aware snapshot rather than a single ranked summary. "–" denotes not reported.

Three observations are useful when reading the table. First, visual-view methods remain highly competitive under standard CD evaluation: strong Layer 1 and Layer 2 systems still achieve high AUC on CDFv2 and retain reasonable transfer to DFDC, which shows that low-level and temporal evidence continue to generalize well when benchmarks emphasize generator shift. Second, Layer 4 methods are already competitive under the same CD setup. RepDFD and LVLMDFD match or exceed lower-layer baselines on CDFv2 and DFDCP, suggesting that language-guided verification can be added without giving up benchmark strength. Third, the table also reveals an evaluation gap for higher layers: Layer 3 is represented here only by an ID result, and many recent multimodal or world-grounded systems still lack standardized CD reporting. This means the table should not be read as evidence that later layers uniformly outperform earlier ones. Rather, it shows that different layers answer different verification needs, while current benchmark practice is still better aligned with artifact and coherence detection than with claim verification or world-level reasoning. Across the CD rows, DFDC also remains the hardest benchmark, with even strong methods mostly landing in the mid-70s to low-80s, which further cautions against naive rank ordering.

5 Evaluation of AIGC-V detection

We provide a comprehensive overview of a generic evaluation framework for AIGC-V detection, focusing on evaluation metrics under the dual-view setting and benchmarks organized in line with three AIGC-V paradigms. We also summarize adjacent non-detector-first diagnostic resources for synthetic-video factual-fidelity evaluation.

5.1 Evaluation Metrics

Shared Basic Metrics. We report standard binary metrics for real-versus-AIGC-V detection, including *Acc*, *AUC*, *Precision*, *Recall*, *F1*, *EER*, and *PR-AUC* under class imbalance. Scores may be computed at the frame level or video level, with temporal aggregation such as mean pooling or voting. These metrics provide a shared baseline, but they do not by themselves diagnose temporal coherence, physical plausibility, or semantic consistency. They can also hide domain-specific score polarization: Cross-AUC explicitly combines per-domain discrimination with prediction-separation stability, showing why averaged dataset-wise AUC can overstate deployment readiness (Nguyen et al., 2026).

Visual View Metrics. The visual view evaluates whether a detector captures perceptual evidence that a video *looks and moves* like a real capture. Two aspects matter. **(i) Intrinsic-cue robustness:** acquisition and generation artifacts such as sensor and ISP traces, resampling and coding footprints, and sampling artifacts are still commonly measured with *Acc*, *AUC*, and *EER* (Cheng et al., 2024; Zhou et al., 2024; Gu et al., 2022b), but the decisive protocols are cross-dataset transfer and perturbation stress tests over codec, bitrate, and resolution. Fixed-operating-point metrics such as *TPR@FPR=α* are also important in deployment-oriented settings. **(ii) Spatiotemporal and physical consistency:** evaluation asks whether motion, layout, and interactions remain temporally and physically plausible (Feng et al., 2023; Zhang et al., 2024a; Anand et al., 2025; Zhang et al., 2025e; Zheng et al., 2025; Kim et al., 2025; Xu et al., 2024). Video-level reporting, temporal-perturbation drops, and motion-ablation studies are generally more informative than strong frame-level scores alone (Kundu et al., 2025; Nie et al., 2024; Yan et al., 2025). A July audit strengthens this caution by tracing apparent gains in several motion detectors to sampling, preprocessing, and real/fake motion-distribution biases (Michels et al., 2026). Overall, visual-view evaluation should prioritize video-level robustness rather than closed-set frame discrimination.

Language View Metrics. The language view evaluates whether video, audio, and accompanying text or metadata jointly support plausible *facts about the world*. It also decomposes into two aspects. **(i) Cross-modal alignment:** benchmarks probe lip-audio synchrony, speaker identity consistency, caption-video matching, and temporally localized mismatches. Depending on the setting, works combine *Acc* and *AUC* with *AP*, *AR*, *Recall@K*, and *mAP* over *IoU* thresholds, often under modality corruption or degraded-channel tests (Katamneni and Rattani, 2024; Xu et al., 2025c; Wang et al., 2026d; Anshul et al., 2025). **(ii) World knowledge and reasoning:** for fact-level and narrative-level verification, standard classification scores are insufficient. Evaluation therefore adds human judgments, pairwise preferences, question answering, and rationale-quality metrics such as *BLEU*, *ROUGE-L*, *METEOR*, *CIDEr*, and embedding-based similarity (Yu et al., 2025; Shen et al., 2025; Wen et al., 2025b; Sun et al., 2025; Gao et al., 2025; Fu et al., 2025; Hondru et al., 2025). Recent studies further show that observers rely on mixed visual, vocal, and knowledge cues, that aggregated crowd judgments can stabilize authenticity screening without reliably recovering missed manipulations, and that detector outputs can mislead downstream claim verification when they are not tied to explicit evidence (Chen and Goh, 2026; Shi et al., 2026; Soprano et al., 2026; Sagar et al., 2026). In short, language-view evaluation shifts the question from "does it look real" to "does it support a plausible, well-grounded story."

What Evaluation Must Show

- **AUC is necessary but insufficient:** it measures separability on a protocol, not whether the decision is evidentially well grounded.
- **Protocols should follow the claimed evidence pathway:** robustness under shift for visual cues, and alignment, localization, grounding, and claim verification with reasoning.
- **Benchmark numbers are not interchangeable.** In-domain accuracy, cross-dataset transfer, operating-point robustness, and explanation quality answer different questions and should not collapse into one headline score.

5.2 Benchmarks

Benchmark development still follows the three AIGC-V paradigms in Section 2, but the three branches now differ not only in scale, but also in annotation granularity and evidential reach. Taken together, these three paradigm-specific benchmark families define the core detector-evaluation landscape. Tables 9–11 summarize this landscape of benchmarks for AIGC-V detection. A complementary line of adjacent diagnostic resources, discussed later in this subsection, broadens that landscape beyond detector-first evaluation toward physical-rule testing, world-dynamics and causality probes, and explanation-oriented diagnosis.

LMV Related Benchmarks. LMV related benchmarks remain the historical backbone of AIGC-V detection because partial edits preserve most of the source video while exposing localized forensic traces. That structure makes LMV especially suitable for testing artifact sensitivity, compression robustness, codec variation, and cross-dataset transfer. Among the foundational resources, FaceForensics++ (Rossler et al., 2019) was decisive because it standardized post-compression manipulation protocols and effectively set the early evaluation culture; Celeb-DF (Li et al., 2020b) and DFDC (Dolhansky et al., 2020) then pushed the field toward harder realism and larger scale; DeeperForensics-1.0 (Jiang et al., 2020) made robustness under real-world distortions a central concern rather than an afterthought. ForgeryNet (He et al., 2021) broadened the manipulation space further. Later resources pushed LMV in more deployment-facing directions: WildDeepfake (Zi et al., 2020) and KoDF (Kwon et al., 2021) moved toward in-the-wild and cross-lingual conditions, AI-Face (Lin et al., 2025b) enlarged demographic coverage, and CDDB (Li et al., 2023) together with DeepfakeBench (Yan et al., 2023) reframed LMV as a protocol problem rather than a single-dataset problem. TalkingHeadBench adds modern talking-head generators and explicit identity- and generator-shift protocols (Xiong et al., 2026b). DD-VQA (Zhang et al., 2024b) and ExDDV (Hondru et al., 2025) begin to attach question answering and explanation supervision to a regime that was originally almost purely artifact-driven, while Beyond Static Artifacts (Gu et al., 2026b) turns temporal deepfake reasoning itself into an explicit benchmark target for VLMs. ActivityForensics, TASLE, and HumanForge extend this branch to activity edits, sparse manipulations in untrimmed streams, and human-interaction forgeries (Bao et al., 2026; Feng et al., 2026b; Xu et al., 2026). ReenactFaces adds a balanced 20K-video facial-reenactment resource designed for cross-dataset evaluation, while DeepfakeImpact weights detector errors by their potential social harm (Petmezas, 2026; Gong et al., 2026).

AVE Related Benchmarks. AVE related benchmarks are organized around whether speech, mouth motion, speaker identity, and language-bearing cues remain jointly coherent over time, so they are less about static artifact capture than about temporal and cross-modal verification. This branch remains smaller than LMV partly because realistic audio-visual edits require synchronized manipulation across modalities together with finer temporal annotation. FakeAVCeleb (Khalid et al., 2021) established the basic paired audio-video manipulation setting, but LAV-DF (Cai et al., 2022) was especially important because it turned localized, content-driven audio-visual mismatch into a first-class benchmark target rather than treating the clip as uniformly fake. Scale then increased with FakeMix (Jung et al., 2024) and AV-Deepfake1M (Cai et al., 2024). The more recent resources differentiate the evaluation question itself: DigiFakeAV (Liu et al., 2025b) targets diffusion-based digital humans, VCapAV (Wang et al., 2025d) matters because it converts AVE into a caption-grounded inconsistency problem, MAVOS-DD (Croitoru et al., 2025) studies multilingual open-set conditions, X-AVFake (Chen et al., 2025a) introduces grounded language reasoning, and MMDF (Kim et al., 2026) broadens manipulation and generator coverage. AVFakeBench further covers both human and general-subject audio-video forgeries, while RAVM evaluates claim, video, audio, and cross-modal misinformation within video-centered cases (Xia et al., 2026; Zhang et al., 2026c). Evaluation in this branch therefore centers on lip-audio alignment, speaker-content consistency, dubbed or spliced segments, temporal localization, and robustness under degraded settings.

GVS Related Benchmarks. GVS related benchmarks are the fastest-moving branch because fully generated videos weaken explicit editing traces while amplifying generator diversity, plausibility failures, and transfer risk. Early resources, including GVF (Ma et al., 2024a), GenVidDet (Ji et al., 2024), GenVideo (Chen et al., 2024a), DVF (Song et al., 2024b), and GenVidBench (Ni et al., 2025), established fully generated video detection as a distinct benchmark regime rather than an extension of face-forgery evaluation. Later benchmarks broadened both scope and supervision: Deepfake-Eval-2024 (Chandra et al., 2025) expanded in-the-wild conditions; GenWorld (Chen et al., 2025c) shifted attention toward real-world simulation; DAVID-X (Gao et al., 2025) and DeeptraceReward (Fu et al., 2025) introduced defect-level or human-perceived trace supervision; and ER-FF++set (Sun et al., 2025) together with AEGIS (Li et al., 2025a) continued the move toward explanation-aware evaluation. The 2026 suites add complementary stressors: Magic Videos preserves native-scale high-fidelity content, ComGenVid supports training-free spatial–temporal likelihood evaluation, CoCoVideo and FVBench broaden generator and manipulation coverage, Artifact-Bench probes diagnostic reasoning, HardGVD emphasizes difficult detail changes, and VidAudit controls evaluation confounds (Li et al., 2026e; Ben Hayun et al., 2026; Feng et al., 2026a; Wang et al., 2026e; Tang et al., 2026; Cheng et al., 2026; Cakiroglu et al., 2026). SafeVid further targets physical, structural, and social-risk reasoning (Wu et al., 2026). AIGVDBench remains particularly consequential because its 440K videos from 31 generators make cross-generator transfer patterns interpretable at ecosystem level (Ma et al., 2026).

What Benchmarks Actually Cover

- **Coverage is structurally uneven.** LMV has the deepest protocol history, AVE remains narrower because synchronized multimodal annotation is costly, and GVS changes fastest as generator turnover keeps resetting the task.
- **The three paradigms test different evidence pathways.** LMV emphasizes localized forensic residue and transfer, AVE temporal and cross-modal coherence, and GVS transfer under weaker artifacts together with plausibility and event logic.
- **Benchmark difficulty shifts with the generation paradigm.** LMV mainly asks whether traces survive compression and domain shift, AVE whether inconsistencies can be localized across modalities, and GVS whether detectors remain reliable as generators become more realistic and diverse.

Table 9 Video-first detection datasets and benchmarks through July 20, 2026: Local Manipulation Video (LMV).

Benchmark & Dataset	Description	Paradigms	Date
A. Local Manipulation Video (LMV)			
FaceForensics++ (Rossler et al., 2019)	Forensics dataset with 1,000 original videos.	*LMV*	01/2019
Celeb-DF (Li et al., 2020b)	Large-scale challenging deepfake-forensics dataset.	*LMV*	09/2019
DeeperForensics-1.0 (Jiang et al., 2020)	Large-scale dataset for real-world face-forgery detection.	*LMV*	05/2020
DFDC (Dolhansky et al., 2020)	Large-scale face-swapping dataset (mostly local forgeries).	*LMV*	06/2020
WildDeepfake (Zi et al., 2020)	In-the-wild dataset for AIGC-V detection.	*LMV*	10/2020
ForgeryNet (He et al., 2021)	Mega-scale benchmark for image and video face-forgery analysis.	*LMV*	06/2021
KoDF (Kwon et al., 2021)	Large-scale Korean AIGC-V detection dataset.	*LMV*	10/2021
CDDB (Li et al., 2023)	Benchmark for easy, hard, and long-sequence AIGC-V detection.	*LMV*	01/2023
DF-Platter (Narayan et al., 2023)	Multi-face heterogeneous deepfake dataset.	*LMV*	06/2023
DeepfakeBench (Yan et al., 2023)	Comprehensive AIGC-V detection benchmark.	*LMV*	07/2023
AI-Face (Lin et al., 2025b)	Million-scale, demographically annotated AI-face dataset.	*LMV&GVS*	06/2024
DD-VQA (Zhang et al., 2024b)	AIGC-V detection VQA (image–question–answer triplets).	*LMV*	09/2024
ExDDV (Hondru et al., 2025)	Benchmark for explainable AIGC-V video detection.	*LMV*	11/2025
FAQ / Beyond Static Artifacts (Gu et al., 2026b)	Forensic benchmark and instruction-tuning resource for temporal video deepfake reasoning in VLMs.	*LMV*	02/2026
ActivityForensics (Bao et al., 2026)	6K+ activity-level forged segments with intra-, cross-domain, and open-world protocols.	*LMV*	04/2026
ReenactFaces (Petmezas, 2026)	10K real and 10K manipulated facial-reenactment videos for cross-dataset evaluation.	*LMV*	04/2026
TASLE (Feng et al., 2026b)	12,472 untrimmed videos with manipulated boundaries, labels, and segment rationales.	*LMV*	06/2026
Localized Streams (Umur et al., 2026)	Short manipulated intervals embedded in otherwise authentic video streams.	*LMV*	06/2026
DeepfakeImpact (Gong et al., 2026)	Two-stage benchmark coupling 12-dataset technical evaluation with social-misjudgment impact.	*LMV*	06/2026
HumanForge (Xu et al., 2026)	18K+ human-centric forged segments with interaction and multi-modal annotations.	*LMV*	07/2026

Table 10 Video-first detection datasets and benchmarks through July 20, 2026: Audio-Visual Editing (AVE).

Benchmark & Dataset	Description	Paradigms	Date
B. Audio-Visual Editing (AVE)			
FakeAVCeleb (Khalid et al., 2021)	Audio-visual multimodal deepfake dataset.	*AVE*	08/2021
LAV-DF (Cai et al., 2022)	Content-driven localized audio-visual deepfake dataset.	*AVE*	11/2022
FakeMix (Jung et al., 2024)	Clip-level benchmark for manipulated audio and video segments.	*AVE*	08/2024
AV-Deepfake1M (Cai et al., 2024)	Large-scale LLM-driven audio-visual deepfake dataset.	*AVE*	10/2024
ArEnAV (Kuckreja et al., 2025)	Audio-visual deepfake dataset for Arabic–English code-switching (CSW).	*AVE*	05/2025
MAVOS-DD (Croitoru et al., 2025)	Multilingual open-set benchmark for audio-visual AIGC-V detection.	*AVE*	05/2025
DigiFakeAV (Liu et al., 2025b)	Benchmark for diffusion-based digital-human audio-visual forgeries.	*AVE*	05/2025
TalkingHeadBench (Xiong et al., 2026b)	Modern talking-head generators with identity- and generator-shift protocols.	*AVE*	05/2025
SocialDF (Batra et al., 2025)	2,126 social-media videos with real and deepfake content under SOTA manipulations.	*AVE*	07/2025
VCapAV (Wang et al., 2025d)	Video-caption-based audio-visual AIGC-V detection dataset.	*AVE*	08/2025
X-AVFake (Chen et al., 2025a)	Dual-modality manipulations with grounded language reasoning.	*AVE*	08/2025
AV-Deepfake1M++ (Cai et al., 2025)	2M clips extending AV-Deepfake1M with diverse audio-visual manipulations.	*AVE*	10/2025
MMDF (Kim et al., 2026)	Multimodal deepfake dataset spanning GAN, diffusion, and flow-matching manipulations.	*AVE*	03/2026
AVFakeBench (Xia et al., 2026)	12K questions over human and general-subject audio-video forgeries.	*AVE*	06/2026
RAVM / VMD-FACT (Zhang et al., 2026c)	Video misinformation with claim, video, audio, and cross-modal manipulations.	*LMV&AVE*	06/2026

Table 11 Video-first detection datasets and benchmarks through July 20, 2026: Generative Video Synthesis (GVS). Broad synthetic-media resources in which video is only a minor modality are outside the core table.

Benchmark & Dataset	Description	Paradigms	Date
C. Generative Video Synthesis (GVS)			
GVF (Ma et al., 2024a)	Generated Video Forensics benchmark for AI-video detection.	*GVS*	02/2024
GenVidDet (Ji et al., 2024)	Real and AIGC-V videos from eight generation models.	*GVS*	05/2024
GenVideo (Chen et al., 2024a)	AIGC-V detection dataset.	*GVS*	05/2024
DVF (Song et al., 2024b)	Diffusion Video Forensics dataset and benchmark.	*GVS*	12/2024
GenVidBench (Ni et al., 2025)	AIGC-V detection dataset.	*GVS*	01/2025
Deepfake-Eval-2024 (Chandra et al., 2025)	Multi-modal in-the-wild benchmark of deepfakes from 2024.	*L&A&G*	03/2025
GenBuster-200K (Wen et al., 2025a)	Real + synthetic videos simulating real-world conditions.	*GVS*	05/2025
GenWorld (Chen et al., 2025c)	Large-scale real-world simulation dataset for AI-video detection.	*GVS*	06/2025
Ivy-Fake (Jiang et al., 2025)	Large-scale benchmark for explainable AIGC detection.	*GVS*	06/2025
DAVID-X (Gao et al., 2025)	AIGC-V with defect-level spatiotemporal annotations + rationales.	*GVS*	06/2025
GenBuster++ (Wen et al., 2025b)	A cross-modal benchmark for VLM evaluation.	*GVS*	07/2025
DeeptraceReward (Fu et al., 2025)	Spatiotemporal benchmark with human-perceived fake traces.	*GVS*	09/2025
ER-FF++set (Sun et al., 2025)	Benchmark with supervision for detection + explanation.	*GVS*	10/2025
AEGIS (Li et al., 2025a)	Large-scale benchmark for AIGC-V authenticity.	*GVS*	10/2025
ViFBench (Li et al., 2026c)	3K videos from 10+ SOTA generators.	*GVS*	12/2025
Video Reality Test (Wang et al., 2025b)	An ASMR-sourced benchmark.	*GVS*	12/2025
AIGVDBench (Ma et al., 2026)	Large-scale benchmark with 440K videos from 31 generation models and 33 evaluated detectors.	*GVS*	01/2026
SynthForensics (Leotta et al., 2026)	Human-centric synthetic-video benchmark with 6,815 videos from five open-source T2V generators.	*GVS*	02/2026
MintVid (Tan et al., 2026)	Lightweight high-quality benchmark with 3K videos from nine state-of-the-art generators.	*GVS*	02/2026
ComGenVid (Ben Hayun et al., 2026)	Recent generators for training-free spatial–temporal likelihood detection.	*GVS*	03/2026
Magic Videos (Li et al., 2026e)	Native-scale benchmark with 140K+ videos from 15 generators and a six-generator test set.	*GVS*	04/2026
SAFE Challenge (Trapeznikov et al., 2026)	6K-video challenge set spanning 13 synthetic-video generators.	*GVS*	05/2026
CoCoVideo (Feng et al., 2026a)	Commercial-model contrastive benchmark for high-quality synthetic video.	*GVS*	05/2026
Artifact-Bench (Tang et al., 2026)	Classification, pairwise realism, and fine-grained artifact diagnosis for MLLMs.	*GVS*	05/2026
FVBench (Wang et al., 2026e)	120K+ real, edited, and generated videos from 42 synthesis/editing models.	*LMV&GVS*	06/2026
HardGVD (Cheng et al., 2026)	Difficult synthetic-video benchmark emphasizing detail and temporal noise changes.	*GVS*	06/2026
VidAudit (Cakiroglu et al., 2026)	Six-control audit protocol, 14-detector toolkit, and confound-aware leaderboard.	*GVS*	06/2026
SafeVid (Wu et al., 2026)	20K videos across 10 social-risk categories with physical and structural reasoning.	*GVS*	07/2026

Diagnostic Resources Beyond Detection Benchmarks. Adjacent diagnostic resources extend evaluation along a clear progression: from basic rule violations, to coherent world dynamics, and finally to explanation-oriented diagnosis. Physical Rule Violations asks whether generated videos obey basic physical constraints: VideoPhy (Bansal et al., 2025), Physics-IQ (Motamed et al., 2025b), and IPV-Bench (Bai et al., 2025) stress physical commonsense and impossible scenarios; Morpheus (Zhang et al., 2025a) and T2VPhysBench (Guo et al., 2025) tighten this into experiment-grounded and first-principles testing; and PhyWorldBench (Gu et al., 2025), VideoPhy-2 (Bansal et al., 2026), and Physion-Eval (Zhang et al., 2026a) move further toward action-centric settings, localized glitches, and expert reasoning traces. World Dynamics and Causality then asks whether those local regularities scale to temporally coherent events and world knowledge over time. WorldSimBench (Qin et al., 2025) makes the world-simulator question explicit; PhyGenBench (Meng et al., 2025) studies multi-step intuitive physics; StoryEval (Wang et al., 2024c) turns consecutive events into a story-level stress test; VideoVerse (Wang et al., 2025e) and T2VWorldBench (Chen et al., 2025d) probe temporal causality, hidden semantics, and world knowledge; and SVBench (Peng et al., 2025) extends the agenda to socially coherent behavior. Controlled analyses (Kang et al., 2024) further argue that this world-model framing should remain a benchmark target rather than an assumed capability. Explanation-Oriented Diagnosis then asks whether a system can turn those failures into explicit, interpretable judgments. SPOTLIGHT (Chinchure et al., 2025) and VideoHallu (Li et al., 2025b) emphasize error localization and hallucination-aware synthetic-video understanding, while TRAVL (Motamed et al., 2025a) and PhyDetEx (Wang et al., 2025f) treat implausibility judgment and explanation as trainable capabilities. Table 12 summarizes this adjacent landscape.

Work and Resource	Description	Type	Date
A. Physical Rule Violations			
VideoPhy (Bansal et al., 2025)	Benchmarks whether generated videos satisfy physical commonsense about objects, attributes, and interactions.	*Eval*	06/2024
Physics-IQ (Motamed et al., 2025b)	Probes whether text-to-video models obey basic physical principles under controlled prompts.	*Eval*	01/2025
IPV-Bench (Impossible Videos) (Bai et al., 2025)	Designs impossible scenarios across physical, geographical, biological, and social domains for stress testing generated videos.	*Eval*	03/2025
Morpheus (Zhang et al., 2025a)	Benchmarks physical reasoning of video generative models with real physical experiments and measurable conservation-law violations.	*Eval*	04/2025
T2VPhysBench (Guo et al., 2025)	Tests first-principles physical consistency in text-to-video generation, including counterfactual robustness and state-transition fidelity.	*Eval*	05/2025
PhyWorldBench (Gu et al., 2025)	Evaluates physical realism of text-to-video models with physics-grounded prompts and anti-physics stress cases.	*Eval*	07/2025
VideoPhy-2 (Bansal et al., 2026)	Extends physical-commonsense evaluation to more challenging action-centric and interaction-heavy generated videos.	*Eval*	01/2026
Physion-Eval (Zhang et al., 2026a)	Provides expert reasoning traces, localized physical glitches, and explanations for physical-realism failures in generated videos.	*Eval*	03/2026
B. World Dynamics and Causality			
WorldSimBench (Qin et al., 2025)	Evaluates whether video generators behave like world simulators by combining perceptual quality and embodied action consistency.	*Eval*	10/2024
Towards World Simulator (PhyGenBench) (Meng et al., 2025)	Crafts a physical-commonsense benchmark for video generation with multi-step dynamics and simulator-style diagnostics.	*Eval*	10/2024
StoryEval (Wang et al., 2024c)	Benchmarks whether T2V models can present short stories composed of consecutive events for future long-video generation.	*Eval*	12/2024
T2VWorldBench (Chen et al., 2025d)	Evaluates world-knowledge generation across physics, nature, activity, culture, causality, and object domains.	*Eval*	07/2025
VideoVerse (Wang et al., 2025e)	World-model-oriented benchmark with hidden semantics, event-level temporal causality, and world-knowledge questions.	*Eval*	10/2025
SVBench (Peng et al., 2025)	Evaluates social reasoning in video generation, including intention, joint attention, norms, and prosocial behavior.	*Eval*	12/2025
C. Explanation-Oriented Diagnosis			
TRAVL (Motamed et al., 2025a)	Adds intra-frame spatial and trajectory-guided temporal attention and fine-tunes on ImplausiBench to improve VLM judgments of physically implausible videos.	*Method*	10/2025
SPOTLIGHT (Chinchure et al., 2025)	Fine-grained identification and localization of generation errors using VLMs.	*Eval*	11/2025
VideoHallu (Li et al., 2025b)	Evaluates multimodal hallucination on synthetic videos and studies mitigation via targeted post-training.	*Eval*	12/2025
PhyDetEx (Wang et al., 2025f)	Introduces a benchmark and trains models to detect and explain violated physical rules in text-to-video outputs.	*Method*	12/2025

Table 12 Adjacent factual-fidelity diagnostic resources relevant to AIGC-V detector design and world-model-oriented evaluation.

6 Challenges and Future

6.1 Robust Diagnostic Evaluation

Chen et al. (2024b) and Sun et al. (2025) argue that classification-centric metrics such as clip-level *AUC*/*EER* are often not sufficiently evidential or interpretable for security-sensitive settings. Under our *factual-fidelity* objective in §3.1, strong closed-set discrimination does not necessarily mean the detector can substantiate *which* real-world proposition is violated, *where* the violation occurs, or *why* the judgment is reliable. Evaluation should therefore be *diagnostic*: it should probe the detector's reliance on complementary pathways (perceptual/temporal cues vs. claim-level semantic verification), and reveal the failure modes that lead to factual-fidelity breakdown.

Concretely, one critical axis is robustness of perceptual and temporal evidence under distribution shift. As diffusion and large video models increasingly suppress noticeable per-frame artifacts, evaluation must stress-test video-level consistency and robustness-to-shift rather than only in-domain performance. This pressure is already visible in explainable detection settings such as EDVD-LLaMA (Sun et al., 2025) and DeeptraceReward (Fu et al., 2025), in generator-diverse detection benchmarks such as GenVidBench (Ni et al., 2025), and even in physics-oriented generative evaluation such as Physics-IQ (Motamed et al., 2025b). Serrano et al. (2026) and Hasan et al. (2026) further show that strong clean-set scores can mask severe brittleness under transfer-based adversarial attacks and modern synthesis pipelines, leaving a wide gap between benchmark discrimination and deployment robustness. Continually refreshed detector snapshots offer one operational response, but their gains must remain versioned and contamination-audited so that temporal adaptation is not mistaken for stable generalization (Miyachi and Uys, 2026). This motivates cross-dataset generalization protocols, systematic perturbation sweeps over codec, bitrate, and resolution, and deployment-oriented reporting at fixed operating points such as *TPR@FPR*=α to better reflect real risk.

A second axis is whether the system can treat a video as making checkable claims and verify them with grounded evidence. Beyond synchronization, cross-modal alignment tests for retrieval and temporal localization should be complemented with fact-level and knowledge-level verification. Common sense reasoning for deepfake detection (Zhang et al., 2024b) pushes evaluation toward explicit question answering, LVLM-DFD (Yu et al., 2025) and AuthGuard (Shen et al., 2025) emphasize grounded multimodal reasoning, and BusterX++ (Wen et al., 2025b) adds unified cross-modal detection-and-explanation settings. These developments also motivate measuring rationale quality and grounding, not just label accuracy. Together, these diagnostics align evaluation with the shift from perceptual-fidelity discrimination to factual-fidelity verification.

6.2 Claim-Level and Dynamic Evaluation

A natural next step is to make benchmarks *evidence-first* by moving from clip-level labels to claim-level supervision. In this setting, each clip is decomposed into a compact set of propositions specifying who, when, where, and what happens, and each proposition is paired with timestamped in-video evidence. Evaluation then scores both claim correctness and evidence grounding, making "what is fake" and "where it is fake" traceable rather than implicit. This direction is consistent with recent explainable AIGC detection resources that augment binary supervision with rationales, traces, and defect-level annotations. X2-DFD (Chen et al., 2024b) and EDVD-LLaMA (Sun et al., 2025) make explanation part of the detector output, Ivy-Fake (Jiang et al., 2025) and DAVID-XR1 (Gao et al., 2025) attach rationales and defect-level traces to AI-generated video analysis, DeeptraceReward (Fu et al., 2025) adds human-perceived fake traces, and ExDDV (Hondru et al., 2025) turns explainable supervision into a benchmark design target.

To prevent shortcut learning and isolate genuine reasoning, evidence-first benchmarks should also incorporate targeted stress tests. One complementary approach is claim-level stress testing for event logic: construct counterfactual or adversarial cases by rewriting event scripts or state transitions while controlling visual-quality confounds, so performance differences primarily reflect relational and event reasoning rather than superficial appearance cues. This concern is explicit in Can We Leave Deepfake Data Behind in Training Deepfake Detector? (Cheng et al., 2024) and FakeChain (Heo and Woo, 2025), both of which expose how easily detectors can rely on shallow cues. In addition, benchmarks increasingly include supervision for tampered locations, spatiotemporal traces, and grounded explanations. EDVD-LLaMA (Sun et al., 2025), DeeptraceReward (Fu et al., 2025), ExDDV (Hondru et al., 2025), and DAVID-XR1 (Gao et al., 2025) all move beyond binary labels in this direction, while Video Reality Test (Wang et al., 2025b) adds realistic high-fidelity content where obvious artifacts are weak. Where available, cross-validating content-side decisions with provenance and authentication signals can further improve trustworthiness in security-sensitive settings.

Finally, given fast-evolving generators, we expect evaluation to move beyond static test sets toward a closed loop between a *real-time detection platform* and a *dynamic benchmark*. In an arena- or leaderboard-style regime, the benchmark is continuously refreshed with outputs from newly released generators and with realistic ingestion or transcoding effects such as codec and resolution changes; detectors are then periodically re-evaluated under a unified protocol to produce regression trajectories rather than one-off scores (Wang et al., 2025b). Such a setup measures not only snapshot accuracy, but also robustness decay, adaptation lag, and failure modes under sustained distribution shift. Hard cases that repeatedly fool strong detectors can then be promoted into subsequent benchmark refreshes with failure-type annotations, turning deployment feedback into iterative stress testing rather than passive reporting.

6.3 Unified Explainable Detection

Wang et al. (2025b) indicates that, for AIGC-V with sparse factual content but high perceptual fidelity and well-aligned modalities, VLMs can perform near random in perceptual-fidelity discrimination, far below humans. This highlights a core gap of language-only detection and the need for visual-view evidence. At the same time, stronger generators and anti-detection pipelines make it increasingly unsafe to assume that authenticity can be decided from perceptual traces alone. Common sense reasoning (Zhang et al., 2024b), generalizable LVLM reasoning (Yu et al., 2025), language guidance (Shen et al., 2025), explainable video reasoning (Gao et al., 2025), and memory-anchored multimodal reasoning (Chen et al., 2025a) all point toward detectors that must reason over semantic and factual content as well. When perceptual fidelity is high and modalities align, decisive evidence may still come from subtle visual statistics or long-range temporal inconsistencies; frequency-aware blending cues (Zhou et al., 2024), local dynamic inconsistency learning (Gu et al., 2022a), natural consistency representations (Zhang et al., 2024a), and plug-and-play video-level blending with spatiotemporal adapter tuning (Yan et al., 2025) all support this view. Conversely, when generators suppress artifacts and preserve visual coherence, visual-only cues can be insufficient and fact-level reasoning becomes necessary; this is exactly the trajectory emphasized by common-sense and reasoning-heavy detection methods (Zhang et al., 2024b; Yu et al., 2025; Shen et al., 2025; Gao et al., 2025). Unified detection is therefore a two-pathway problem: perceptual evidence plus fact-level verification.

In practice, this motivates cross-layer evidence fusion: frame-level intrinsic cues from Layer 1, sequence-level motion and physics coherence from Layer 2, multimodal verification from Layer 3, and external-knowledge reasoning from Layer 4 should be integrated into a unified evidence graph, where corroborating evidence boosts confidence and conflicts trigger abstention or human review. The goal, however, is not to average heterogeneous scores, but to preserve evidential granularity so that low-level traces, localized cross-modal conflicts, and claim-level checks remain individually inspectable. The key challenge is to connect low-level cues to high-level conclusions in an auditable way, as X2-DFD (Chen et al., 2024b) and EDVD-LLaMA (Sun et al., 2025) already make clear. A practical system should output structured evidence objects, such as suspicious segments and defect-level traces (Gao et al., 2025; Anshul et al., 2025), localized cross-modal mismatches (Katamneni and Rattani, 2024; Chen et al., 2024b), and tested claims or entities (Zhang et al., 2024b), rather than only free-form rationales. Tool-augmented pipelines such as LAVID (Liu et al., 2025d) help only when each tool invocation is tied to a concrete sub-claim and yields reusable evidence objects; memory-anchored multimodal reasoning (Chen et al., 2025a) points to the same requirement.

6.4 Evidence-First Trustworthy Detection

To match trustworthy detection that truly targets factual fidelity, system design should follow an Evidence-First principle with an explicit reasoning path: Identify, Localize, and Explain. X2-DFD (Chen et al., 2024b), DeeptraceReward (Fu et al., 2025), and EDVD-LLaMA (Sun et al., 2025) already move in this direction. The key implication is that explanation should be downstream of evidence extraction, not a post-hoc verbal gloss over a classifier score. This path should separate candidate-issue discovery, where-and-when localization, and explanation of which perceptual, temporal, cross-modal, or fact-level constraints are violated. Such a design requires consistent evidence schemas, as DAVID-XR1 (Gao et al., 2025) and X2-DFD (Chen et al., 2024b) suggest, together with calibrated uncertainty emphasized by reliability-centered analyses (Wang et al., 2024a). Calibration matters because trustworthy deployment depends on knowing when evidence is weak, conflicting, or incomplete rather than forcing a binary answer.

Within this dual-pathway setup, each tool invocation or knowledge citation should be bound to a specific argumentation step (Liu et al., 2025d). Content-side analysis should also be cross-validated against source-side provenance and authentication signals when available, rather than treated as an unrelated defense channel. This connection is explicit in media-forensics and defense surveys (Verdoliva, 2020; Deng et al., 2025), in technical provenance standards such as C2PA (Coalition for Content Provenance and Authenticity, 2024), and in complementary verification frameworks built on face geometry (Tursman et al., 2020) or blockchain-backed authenticity pipelines (Hajjej et al., 2026). This matters beyond standalone detection accuracy: Sagar et al. (2026) suggests that detector outputs can become misleading priors when they are injected into claim verification pipelines without explicit evidence grounding. The practical difficulty is to reconcile potentially conflicting evidence and surface it in a unified evidence space that supports auditing and abstention under uncertainty (Wang et al., 2024a). A trustworthy system should therefore preserve both agreement and disagreement across evidence sources, rather than collapsing them too early into a single confidence score.

What Trustworthy Detection Requires

- **Diagnose the real failure.** Evaluation should reveal where factual fidelity breaks, not only whether real and fake clips separate on a closed set.
- **Supervise evidence, not only labels.** Claim-level annotations, timestamped evidence, shortcut controls, and dynamic testbeds are needed as generators evolve.
- **Return structured evidence, not only scores.** Cross-layer fusion is useful only when cues become traceable claims, segments, entities, or mismatches rather than free-form confidences or rationales.
- **Treat deployment as evidence governance.** Provenance checks, calibration, and abstention are part of trustworthy detection when evidence conflicts or remains incomplete.

7 Related Work

Table 13 systematically compares representative surveys by what they take as the task, what deployment concerns they foreground, and how far they move from artifact-centric detection toward grounded verification. This comparison clarifies that our contribution is not only broader coverage, but also a different organizing principle for AIGC-V detection.

7.1 Scope and Framing

Video-targeted scope. Most prior surveys take either a face-manipulation view or a broad *deepfake* or AIGC scope, so they naturally organize the field around generation pipelines, detector families, and generic forensic cues. This breadth is useful for coverage, but it often compresses important distinctions among local manipulation, audio-visual editing, and fully generated video, even though these regimes differ in threat model, evidence availability, and evaluation difficulty. Broad overviews by Tolosana et al. (2020), Mirsky and Lee (2021), and Nguyen et al. (2022) are representative of this framing. From a media-forensics viewpoint, Verdoliva (2020) places deepfakes within a larger integrity-verification pipeline of detection, localization, attribution, and authentication. More video-oriented surveys, including Hashmi et al. (2024) on audio-visual deepfakes and

Survey	Scope & Framing			Language View		
	Video-Targeted Scope	Trust-worthiness Framing	Factual-Fidelity Verif.	Explain-ability & Loc.	Semantic Cues	World-level Reasoning
Wang et al. (2024a)	✗	✓	✗	✗	✗	✗
Xu et al. (2025a)	✗	✓	✗	✗	✗	✗
Croitoru et al. (2024)	✗	✓	✗	✗	✗	✗
Hashmi et al. (2024)	✓	✗	✗	✗	✗	✗
Kaur et al. (2024)	✓	✓	✗	✗	✗	✗
Deng et al. (2025)	✗	✓	✗	✓	✗	✗
Nguyen-Le et al. (2025)	✗	✓	✗	✓	✗	✗
Lin et al. (2024a)	✗	✓	✗	✓	✓	✗
Zou et al. (2025)	✗	✓	✗	✓	✓	✗
Ours	✓	✓	✓	✓	✓	✓

Table 13 Comparison of representative surveys and ours across (i) **Scope and Framing** (video-targeted scope; trustworthiness framing; factual-fidelity verification) and (ii) **Language View** (explainability and localization; semantic cues; world-level reasoning). ✓ indicates substantial discussion; ✗ indicates absent or brief mention.

Kaur et al. (2024) on video-specific challenges, move closer to our focus, but they still do not systematize the full AIGC-V landscape across the three paradigms used here.

Trustworthiness framing. Recent surveys increasingly move beyond raw accuracy. Wang et al. (2024a) and Deng et al. (2025) shift attention toward deployment reliability, adversarial evolution, uncertainty, failure modes, and complementary defenses such as authentication and disruption. Beyond the deepfake-only scope, Lin et al. (2024a) surveys AI-generated multimedia across modalities and explicitly separates better classification from broader goals such as generalizability, robustness, and interpretability. Zou et al. (2025) further highlights the transition from domain-specific detectors to VLM- and MLLM-based systems with localization and explanation. Yet even under this framing, trustworthiness is still usually defined around making detector outputs more reliable, rather than around making the underlying authenticity judgment explicitly evidential and claim grounded.

Factual-fidelity verification. Accordingly, "verification" in prior surveys usually appears either inside a broader media-forensics workflow (Verdoliva, 2020) or inside a deployment-reliability discussion centered on robustness, uncertainty, and countermeasures (Wang et al., 2024a; Deng et al., 2025). Both perspectives are important, but neither reorganizes the task around what a video claims and whether those claims remain consistent with the world. Lin et al. (2024a) and Zou et al. (2025) bring in semantic cues and interpretability with localization, but the emphasis still falls on recognizing manipulations, localizing suspicious regions, or justifying predictions. Fact-level checking of who did what, where, and when is rarely treated as the primary unit of analysis. As summarized in Table 13, this is the main framing gap that motivates our survey.

7.2 Language View

Explainability & localization. Recent surveys increasingly recognize explainability and localization, especially in trustworthy and VLM-based detection. Lin et al. (2024a), Zou et al. (2025), Deng et al. (2025), and Nguyen-Le et al. (2025) all acknowledge that deployment requires more than a scalar real/fake score. However, explanation is still usually understood as saliency, manipulated-region localization, or natural-language justification attached to a prediction. What remains less explicit is whether those outputs become reusable evidence objects that can support later claim verification and auditing.

Semantic cues. As deepfakes move from static artifacts to temporally coherent synthesis, surveys increasingly emphasize video-level evidence and multimodal cues beyond pixels. Kaur et al. (2024) highlights video-specific challenges and robustness under real-world degradations, while Hashmi et al. (2024) centers audio-visual correspondence, identity preservation, and synchrony cues. Lin et al. (2024a) and Zou et al. (2025) push further by acknowledging semantic cues in language-enabled detectors, and Nguyen-Le et al. (2025) together with Xu et al. (2025a) consolidate passive detection cues across modalities and emerging AIGC forensics directions. Even so, semantic information is usually introduced as auxiliary context that improves classification or explanation, not as a primary evidential pathway for testing factual fidelity.

World-level reasoning. Explicit world-level reasoning about events, causality, and plausibility remains even less developed in the survey literature. Most prior surveys stop at semantic alignment or multimodal consistency and do not elevate world knowledge, event logic, or external verification into a first-class evidence pathway. Table 13 makes this asymmetry clear: semantic cues are beginning to appear, but world-level reasoning and, especially, fact-level factual-fidelity verification remain under-emphasized. Our survey addresses this gap by adopting a video-targeted scope, foregrounding factual-fidelity verification, and organizing methods, evaluation, and benchmarks under our taxonomy.

8 Conclusion

This survey reframes AI-generated video detection as factual-fidelity verification and synthesizes prior work through a Vision-Language Dual-View four-layer taxonomy with aligned benchmarks and metrics. Beyond summarizing methods, we connect these layers to deployment challenges, evaluation gaps, and emerging trends, and distill key requirements for trustworthy detection, including evidence-first and traceable decisions as well as robustness across generators and real-world scenarios. We expect this survey to provide a clear and actionable reference for future AIGC-V detection research and practice.

More broadly, trustworthy AIGC-V detection should become a shared agenda for the computer vision, natural language processing, multimodal, and world model communities. CV contributes spatiotemporal evidence and forensic robustness; NLP contributes decomposition, reasoning, grounding, and explanation; multimodal and world model research contribute stronger cross-modal alignment and richer priors about physical, causal, and temporal coherence. Together, these ingredients can push detection beyond artifact hunting toward a stricter notion of realism: not whether a video merely looks plausible, but whether its entities, events, and dynamics remain faithful to the constraints of the real world.

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